Cortex Tokenomics: A Game-Theoretic Framework for Sustainable Value Accrual in Autonomous Agent Economies

David Garcia
PureCortex Research
research@purecortex.io

March 2026

Abstract

We present a comprehensive tokenomic framework for *PureCortex*, an autonomous agent economy built on the Algorand blockchain. The framework comprises six interlocking pillars: (i) quadratic bonding curve graduation with creator vesting, (ii) volatility-adaptive dynamic fees, (iii) vote-escrowed staking (veCortex) with day-one launch parity, (iv) protocol-level buyback-and-burn mechanics, (v) composability rewards via cross-agent tool invocation, and (vi) sovereign on-chain governance anchored by a constitutional AI Senator. We derive closed-form expressions for bonding curve cost, dynamic fee revenue, staking yield, and deflationary crossover conditions. Game-theoretic analysis demonstrates that the mechanism admits a unique Nash equilibrium in which rational agents stake rather than sell, creators commit to long-horizon vesting, and governance delegation is individually rational when information acquisition costs exceed voting costs. Monte Carlo simulations across 1,000 trajectories per scenario confirm a deflationary crossover at approximately month 18–24, staker yields in the 15–40% APR range, and superlinear network value growth consistent with Metcalfe's law. A 14-dimension comparative analysis against existing agent tokenization platforms reveals structural advantages across creator incentives, fee adaptation, composability, and governance. The full constitutional text, contract ABIs, and simulation methodology are provided in the appendices.

Contents

1 Introduction **6**
1.1 Motivation . . . 6
1.2 Problem Statement . . . 6
1.3 Contributions . . . 7
1.4 Paper Organization . . . 7

2 Related Work **8**
2.1 Bonding Curves and Continuous Token Models . . . 8
2.2 Vote-Escrowed Tokenomics . . . 8
2.3 Agent Tokenization Platforms . . . 8
2.4 Game Theory and Mechanism Design in DeFi . . . 9
2.5 Algorand as Settlement Layer . . . 9

3 Token Architecture **9**
3.1 Token Parameters . . . 10
3.2 Allocation Structure . . . 10
3.3 Vesting Formulas . . . 10
3.3.1 Creator Vesting . . . 11
3.3.2 Staking Emissions . . . 11
3.3.3 Genesis Distribution . . . 11
3.4 Smart Contract Architecture . . . 11

4 Bonding Curve Graduation **12**
4.1 Quadratic Pricing Function . . . 12
4.2 Cost Function . . . 13
4.3 Monotonicity and Convexity . . . 13
4.4 Graduation Mechanism . . . 14
4.4.1 LP Lock via LogicSig . . . 14
4.5 Graduation Bonus Distribution . . . 14
4.6 Comparison of Curve Shapes . . . 15

5 Dynamic Fee Mechanism **15**
5.1 Fee Function . . . 16
5.2 Protocol Revenue Split . . . 16
5.3 Revenue Superiority of Dynamic Fees . . . 16
5.4 Adverse Selection Reduction . . . 17
5.5 On-Chain Volatility Oracle . . . 18

6 Vote-Escrowed Staking (veCortex) **18**
6.1 Lock Mechanics . . . 18
6.2 Linear Decay . . . 19
6.3 Emission Distribution . . . 19
6.4 Boost Multiplier . . . 19
6.5 Day-One Launch Advantage . . . 19
6.6 Cumulative Reward Analysis . . . 20
6.7 Staking Equilibrium . . . 20

7 Buyback-and-Burn Mechanism **21**
7.1 The Assistance Fund . . . 21
7.2 Revenue Sources . . . 21
7.3 Continuous Buyback Mechanism . . . 21
7.4 Circulating Supply Model . . . 22
7.5 Deflationary Crossover . . . 22
7.6 Crossover Timing Analysis . . . 23
7.7 Long-Run Supply Convergence . . . 23

8 Composability Rewards **24**
8.1 Cross-Agent Tool Invocation . . . 24
8.2 x402 Micropayment Flow . . . 24
8.3 Composability Score . . . 25
8.4 Network Value Dynamics . . . 25

8.5 Anti-Sybil Properties . . . 25
8.6 Composability as Moat . . . 26

9 Sovereign Governance **26**
9.1 Senator AI Architecture . . . 26
9.2 Lawmaker Agents . . . 27
9.3 Delegation Model . . . 27
9.4 Proposal Lifecycle . . . 27
9.5 Proposal Types . . . 28
9.6 Constitutional Framework . . . 28
9.6.1 Immutable Preamble . . . 28
9.6.2 Amendable Articles . . . 29
9.7 Senator Welfare Function . . . 29

10 Game-Theoretic Analysis **29**
10.1 Stake vs. Sell Game . . . 30
10.2 Repeated Game and the Folk Theorem . . . 31
10.3 Creator Commitment . . . 31
10.4 Bonding Curve as Quality Signal . . . 32
10.5 Senator Mechanism Design . . . 32
10.6 Delegation Game . . . 33
10.7 Constitutional Rigidity as Schelling Point . . . 33

11 Simulations **34**
11.1 Methodology . . . 34
11.2 Supply Trajectory . . . 34
11.3 Bonding Curve Comparison . . . 35
11.4 Staker Yield by Lock Duration . . . 36
11.5 Dynamic vs. Flat Fee Revenue . . . 37
11.6 Stress Test: Crash Recovery . . . 38
11.7 Network Value Growth . . . 39
11.8 Governance Simulation . . . 40
11.9 Sensitivity Analysis . . . 41

12 Security Analysis **42**
12.1 Integer Precision in Bonding Curve Arithmetic . . . 42
12.1.1 Truncation Bug . . . 42
12.1.2 Fix: Scaled Arithmetic . . . 42
12.2 Reentrancy Elimination via Atomic Transactions . . . 42
12.3 LogicSig LP Lock Immutability . . . 43
12.4 Oracle-Free Design . . . 43
12.5 Contract Sovereignty . . . 44
12.6 Dual-Brain Governance Safety . . . 44
12.7 Attack Surface Summary . . . 45

13 Comparative Analysis **45**
13.1 Comparison Framework . . . 45
13.2 Detailed Analysis by Dimension . . . 46
13.2.1 Creator Vesting . . . 46
13.2.2 Staking Timing . . . 47
13.2.3 Fee Model . . . 47
13.2.4 Revenue to Holders . . . 47
13.2.5 Burn Mechanics . . . 47
13.2.6 Composability . . . 48
13.2.7 Governance . . . 48
13.2.8 Chain Properties . . . 48
13.3 Quantitative Impact Summary . . . 48

14 Conclusion **49**
14.1 Summary of Results . . . 49
14.2 Key Quantitative Results . . . 49
14.3 Limitations . . . 50
14.4 Future Work . . . 50

A Full Proofs **51**
A.1 Proof of Proposition 4.2: Monotonicity and Convexity . . . 51
A.2 Proof of Theorem 4.5: Quadratic vs. Linear vs. Exponential . . . 51
A.3 Proof of Theorem 10.3: Cooperation via Lock Commitment . . . 53
A.4 Proof of Proposition 10.4: Vesting Maximizes Expected Wealth . . . 54
A.5 Proof of Theorem 10.6: Separating Equilibrium . . . 54
A.6 Proof of Theorem 10.7: Pareto Improvement via Governance . . . 55
A.7 Proof of Proposition 10.8: Rationality of Delegation . . . 56

B Simulation Methodology **56**
B.1 Simulation Engine . . . 56
B.2 Parameter Tables . . . 56
B.3 Volatility Model . . . 58
B.4 Trading Volume Model . . . 58
B.5 Composability Model . . . 58
B.6 Governance Model . . . 58
B.7 Random Seed Generation . . . 59
B.8 Stress Test Methodology . . . 59
B.9 Confidence Interval Construction . . . 59

C Contract ABIs **59**
C.1 AgentFactory Contract . . . 59
C.1.1 State Schema . . . 60
C.1.2 ARC-4 Methods . . . 60
C.2 SovereignTreasury Contract . . . 62
C.2.1 State Schema . . . 62
C.2.2 ARC-4 Methods . . . 62
C.3 veCORTEX Contract . . . 63
C.3.1 State Schema . . . 64

C.3.2 ARC-4 Methods 64

D PureCortex Constitution **66**

1 Introduction

The emergence of large language models and autonomous AI agents has catalyzed a paradigm shift in decentralized infrastructure. Where blockchains once served primarily as ledgers for value transfer and smart contract execution, they now function as *coordination substrates* for economically sovereign AI agents—entities capable of invoking tools, managing treasuries, and transacting with both humans and other agents without centralized intermediation. This transformation demands tokenomic architectures purpose-built for agent economies, yet the prevailing designs inherit assumptions from human-centric decentralized finance (DeFi) that fail under the distinct incentive landscape of autonomous actors.

1.1 Motivation

An AI agent operating within an on-chain economy requires infrastructure that simultaneously satisfies five desiderata:

1. **Capital formation.** Agents must raise initial capital through mechanisms that price discovery fairly and reward early supporters without enabling manipulative extraction.
2. **Revenue sustainability.** The protocol must generate fee revenue that adapts to market conditions, ensuring solvency during low-activity periods without extracting punitive rents during calm markets.
3. **Stakeholder alignment.** Creators, stakers, users, and the protocol treasury must share revenue in proportions that sustain participation across all roles.
4. **Composability.** Agents must be able to invoke one another's capabilities (tools, data feeds, inference endpoints) through standardized protocols, with economic incentives that reward interoperability.
5. **Governance.** Protocol parameters, treasury allocations, and constitutional norms must be governable by stakeholders, with safeguards against both capture and ossification.

1.2 Problem Statement

Existing agent tokenization platforms exhibit five structural deficiencies that undermine long-term ecosystem health:

P1: No creator vesting. Creators receive their full token allocation at launch, enabling immediate liquidation. This misaligns creator incentives with long-term agent development and facilitates pump-and-dump dynamics wherein creators abandon agents after extracting initial liquidity.

P2: Post-hoc staking disadvantage. When staking mechanisms are introduced after token launch, early adopters who provided initial liquidity and bore maximum risk receive no preferential treatment. Late entrants capture identical yields despite contributing no price-discovery capital.

P3: Flat fee structures. A fixed percentage trading fee (e.g., 1%) fails to adapt to volatility regimes. During high-volatility periods, flat fees are insufficient to compensate liquidity providers for adverse selection; during low-volatility periods, they impose unnecessary friction that suppresses trading volume.

P4: Agent silos. Without standardized inter-agent communication protocols and economic incentives for composability, each agent operates as an isolated service. This prevents the emergence of compound agent workflows—multi-agent pipelines where specialized agents collaborate on complex tasks.

P5: No on-chain governance. Protocol parameters are controlled by centralized teams with no formal governance mechanism, no constitutional framework, and no stakeholder recourse. This introduces platform risk and regulatory uncertainty.

1.3 Contributions

We present a six-pillar tokenomic framework that addresses each deficiency:

1. **Bonding curve graduation with creator vesting** (Section 4). A quadratic bonding curve provides transparent price discovery during agent launch, with mandatory 180-day creator vesting that releases 1% immediately and the remainder linearly. Upon reaching a graduation threshold, liquidity atomically migrates to a decentralized exchange with a 10-year LP lock.

2. **Volatility-adaptive dynamic fees** (Section 5). A continuous fee function maps realized volatility to trading fees in the range $[0.5\%, 2.0\%]$, with a five-way revenue split among stakers, creators, treasury, referrers, and composability rewards.

3. **Vote-escrowed staking** (Section 6). The veCortex mechanism, available from day one of platform launch, grants governance weight and revenue share proportional to both the quantity staked and the lock duration, with a maximum four-year commitment.

4. **Protocol-level buyback-and-burn** (Section 7). Multiple revenue streams fund a weekly buyback with a supply-proportional cap, establishing conditions under which the token supply becomes net deflationary.

5. **Composability rewards** (Section 8). Cross-agent tool invocation via the Model Context Protocol (MCP) generates micropayment flows with a composability tax, and a Composability Score distributes weekly rewards to the most-integrated agents.

6. **Sovereign on-chain governance** (Section 9). A constitutional governance framework features a protocol-owned AI Senator that proposes but cannot vote, veCortex-weighted Lawmaker Agents with liquid delegation, and a multi-tiered proposal lifecycle anchored by an on-chain constitution.

We provide formal game-theoretic analysis (Section 10) demonstrating that the mechanism admits equilibria aligned with ecosystem health, Monte Carlo simulations (Section 11) validating quantitative predictions, security analysis (Section 12), and a 14-dimension comparative analysis (Section 13) against existing platforms.

1.4 Paper Organization

Section 2 surveys related work in bonding curves, vote-escrowed tokenomics, agent tokenization, and game-theoretic mechanism design. Section 3 presents the overall token architecture, allocation, and vesting schedules. Sections 4 through 9 detail each of the six pillars. Section 10 provides formal game-theoretic analysis. Section 11 presents simulation results. Section 12 addresses

security considerations. Section 13 provides comparative analysis. Section 14 concludes. The appendices contain full proofs (Appendix A), simulation methodology (Appendix B), contract ABIs (Appendix C), and the full constitutional text (Appendix D).

2 Related Work

This section surveys four streams of literature that inform the PureCortex tokenomic design: bonding curves and automated market makers, vote-escrowed tokenomics, agent tokenization platforms, and game-theoretic mechanism design for decentralized systems.

2.1 Bonding Curves and Continuous Token Models

The concept of automated price discovery through mathematical functions originates with Hanson's logarithmic market scoring rule for prediction markets [9]. Hanson demonstrated that a cost function derived from a convex potential can simultaneously serve as a market maker and an information aggregation device. The Bancor protocol generalized this insight to arbitrary token issuance, introducing continuous liquidity through a reserve-backed bonding curve parameterized by a connector weight [10]. Subsequent work by Angeris et al. [1] provided rigorous analysis of constant-product market makers (the Uniswap model), establishing conditions under which arbitrageurs drive prices toward external reference values.

Our quadratic bonding curve (Section 4) builds on this lineage but introduces two innovations absent from prior work: a deterministic graduation mechanism that atomically transitions liquidity from the bonding curve to a decentralized exchange, and a mandatory creator vesting schedule that conditions token release on elapsed time rather than price milestones.

2.2 Vote-Escrowed Tokenomics

The vote-escrowed (ve) model was pioneered by Curve Finance's veCRV [4, 6], wherein users lock CRV tokens for up to four years to receive governance weight proportional to the remaining lock duration. This mechanism simultaneously reduces circulating supply, aligns governance power with long-term commitment, and distributes protocol revenue to lockers. Balancer subsequently adapted the model as veBAL [2], adding a gauge-based emission allocation system.

Two limitations of existing ve implementations motivate our design. First, veCRV was introduced approximately two years after CRV launch, disadvantaging early participants who could not lock tokens from inception. Second, the linear decay of governance weight creates a "cliff problem" wherein users must repeatedly re-lock to maintain influence, introducing gas costs and attentional overhead. The veCortex mechanism (Section 6) addresses the first limitation by launching contemporaneously with the token, and mitigates the second through a boost multiplier that rewards sustained participation.

2.3 Agent Tokenization Platforms

The tokenization of AI agents—the process of creating tradeable digital assets representing ownership stakes in or usage rights to autonomous agents—has emerged as a distinct category within decentralized infrastructure. The most prominent exemplar, which we refer to as *Platform A* for neutrality, implements the following design:

- A fixed supply of 10^9 tokens per agent, with 1% allocated to the creator at launch and no vesting schedule.

- A flat 1% trading fee on all buy and sell transactions, split between the creator and the protocol.
- Initial liquidity bootstrapped through a bonding curve, with LP tokens locked upon migration to a decentralized exchange.
- No on-chain governance mechanism; protocol parameters are controlled by the development team.
- No composability framework; agents operate as isolated services.

While Platform A demonstrated product-market fit and catalyzed the agent token category, its design exhibits the five structural deficiencies enumerated in Section 1.2. The absence of creator vesting has been empirically associated with rapid creator exits; the flat fee structure fails to adapt to volatility regimes; and the lack of governance creates platform risk.

2.4 Game Theory and Mechanism Design in DeFi

Roughgarden [13] established a formal framework for analyzing transaction fee mechanisms as games between users and block producers, proving that EIP-1559-style mechanisms are incentive-compatible under certain conditions. Daian et al. [5] identified miner extractable value (MEV) as a fundamental game-theoretic challenge in decentralized exchanges, demonstrating that frontrunning and sandwich attacks constitute rational strategies under existing fee structures.

In the broader mechanism design literature, Myerson [12] established revenue-optimal auction theory, while Spence [14] introduced the signaling model that we adapt in Section 10 to analyze bonding curve participation as a quality signal. Zargham et al. [16] proposed cadCAD-based simulation frameworks for economic games in token ecosystems, providing methodological grounding for our simulation approach.

The intersection of game theory and governance has been explored in the context of liquid democracy and delegative voting [7]. Our governance model (Section 9) extends this literature by introducing an AI Senator agent that participates in the proposal process but is constitutionally prohibited from voting, creating a novel separation of powers within a token-governed system.

2.5 Algorand as Settlement Layer

The Algorand blockchain [3, 8] provides several properties that are structurally advantageous for agent tokenomics: immediate transaction finality (no probabilistic confirmation), native asset support through the Algorand Standard Asset (ASA) model, atomic transaction groups that enable multi-step operations without reentrancy risk, and LogicSig-based programmable accounts that can enforce time-locked LP positions without trusted intermediaries.

These properties eliminate entire classes of vulnerabilities present in EVM-based implementations (see Section 12) and enable the atomic graduation mechanism described in Section 4.

3 Token Architecture

This section specifies the CORTEX token parameters, allocation structure, vesting schedules, and smart contract architecture that form the foundation of the PureCortex economy.

3.1 Token Parameters

Table 1 summarizes the core token parameters.

Table 1: CORTEX token parameters.

Parameter	**Value**
Name	CORTEX
Standard	Algorand Standard Asset (ASA)
Total Supply	10,000,000,000,000,000 (10^{16}, i.e., 10 quadrillion)
Decimals	6
Smallest Unit	10^{-6} CORTEX (1 μCORTEX)
Chain	Algorand MainNet
Finality	Instant ($<$ 3.3 s block time)
Transferability	Unrestricted after TGE

The large total supply (10^{16}) with six decimal places ensures sufficient granularity for micropayment flows in agent-to-agent transactions (see Section 8) while maintaining human-readable unit prices during early-stage price discovery.

3.2 Allocation Structure

The total supply is partitioned into six categories as specified in Table 2. The allocation reflects a community-first philosophy: **0% to VCs, 0% to team/advisors** — only a single creator allocation of 10%, with 90% directed to community and ecosystem participants.

Table 2: CORTEX token allocation.

Category	**Percentage**	**Tokens ($\times 10^{12}$)**	**Vesting**
Creator	10%	1,000	10% at TGE, 90% daily over 180 days
Genesis Distribution	31%	3,100	Airdrop to early testnet users & Algorand comm
Future Emissions & Staking	24%	2,400	veCORTEX rewards, 4-year halving decay
Liquidity	15%	1,500	CORTEX/ALGO DEX LP (10-year lock) + age
Agent Incentives	15%	1,500	Graduation bonuses, composability rewards, per
Assistance Fund	5%	500	Seed for buyback engine; self-sustaining from re
Total	**100%**	**10,000**	

The Assistance Fund receives 90% of *all* protocol revenue (trading fees, agent creation fees, x402 commissions, LP fees) and uses it to continuously buy CORTEX from the open market and burn it. The initial 5% seed provides the fund with operational capital at launch; thereafter it is entirely self-sustaining from protocol revenue. The remaining 10% of protocol revenue covers operations and development (see Section 5).

3.3 Vesting Formulas

We define three distinct vesting schedules, each with a closed-form expression.

3.3.1 Creator Vesting

The creator allocation ($A_c = 10\%$ of total supply) vests according to:

$$V_{\text{creator}}(d) = 0.10 \cdot A_c + 0.90 \cdot A_c \cdot \frac{\min(d, 180)}{180} \tag{1}$$

This releases 10% of the allocation at the Token Generation Event (enabling the creator to provide initial liquidity and demonstrate commitment) and vests the remaining 90% linearly over 180 days via daily unlocks. The creator's unvested tokens are held in a smart contract escrow; premature withdrawal attempts revert. There is no separate team or advisor allocation — the creator allocation is the sole insider distribution.

3.3.2 Staking Emissions

The staking emission pool of $R_{\text{total}} = 2.4 \times 10^{15}$ tokens (24% of total supply) is distributed over a four-year period with declining annual weights:

$$E(y) = R_{\text{total}} \cdot \frac{w_y}{\sum_{i=1}^{4} w_i} \tag{2}$$

where the weight vector is $\mathbf{w} = (0.4,\ 0.3,\ 0.2,\ 0.1)$ and $y \in \{1, 2, 3, 4\}$ denotes the year. This yields:

$$E(1) = 2.4 \times 10^{15} \times 0.4 = 9.6 \times 10^{14} \tag{3}$$
$$E(2) = 2.4 \times 10^{15} \times 0.3 = 7.2 \times 10^{14} \tag{4}$$
$$E(3) = 2.4 \times 10^{15} \times 0.2 = 4.8 \times 10^{14} \tag{5}$$
$$E(4) = 2.4 \times 10^{15} \times 0.1 = 2.4 \times 10^{14} \tag{6}$$

The declining schedule front-loads rewards to incentivize early staking while ensuring four full years of emission-funded yield.

3.3.3 Genesis Distribution

The genesis distribution of $A_{\text{gen}} = 3.1 \times 10^{15}$ tokens (31% of total supply) is released in full at TGE via airdrop to early testnet participants and the Algorand community. There is no vesting schedule for this allocation — recipients have immediate, unrestricted access:

$$V_{\text{gen}}(d) = A_{\text{gen}} \quad \forall\, d \geq 0 \tag{7}$$

The immediate release ensures broad distribution from day one and maximizes the decentralization of the circulating supply at launch.

3.4 Smart Contract Architecture

The PureCortex protocol is implemented as a system of three interacting Algorand smart contracts, depicted schematically in Figure 1.

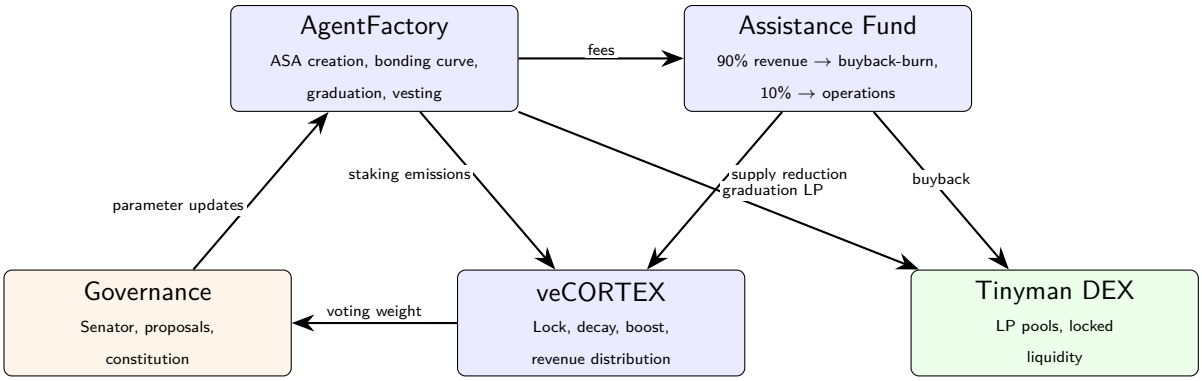

Figure 1: Smart contract architecture. Arrows indicate primary data and value flows between contracts. Tinyman DEX and Governance are external to the core protocol but interact through well-defined interfaces.

AgentFactory. Handles agent creation, bonding curve pricing, token minting via ASA inner transactions, creator vesting escrow, and graduation (atomic LP formation on Tinyman). Stores per-agent state in Algorand box storage: current supply, base price, slope, graduation status, and creator vesting schedule.

SovereignTreasury (Assistance Fund). Receives 90% of all protocol revenue from trading fees, agent creation fees, x402 commissions, and LP fees. Executes continuous buyback-and-burn operations (Section 7) by purchasing CORTEX from the open DEX market and sending it to the Algorand zero address. The remaining 10% of protocol revenue is directed to an operations account for development and infrastructure costs. Seeded with 5% of total supply at genesis; thereafter entirely self-sustaining from protocol revenue.

veCORTEX. Manages token locking, computes time-decaying governance weight, and calculates boost multipliers for staking emissions. Lock state is stored in per-user box storage with fields for locked amount, lock duration, lock start timestamp, and accumulated emission rewards. Fee revenue no longer flows directly to stakers — instead, stakers benefit from the continuous supply reduction executed by the Assistance Fund.

All three contracts interact through Algorand atomic transaction groups, ensuring that multi-step operations (e.g., graduation: sell remaining bonding curve supply, form LP pair, lock LP tokens, distribute graduation bonus) either complete entirely or revert entirely. This atomicity eliminates the reentrancy and partial-execution risks inherent in EVM-based designs.

4 Bonding Curve Graduation

This section formalizes the quadratic bonding curve used for agent token issuance, derives the cost function, proves key properties, specifies the graduation mechanism, and compares the quadratic form against alternative curve shapes.

4.1 Quadratic Pricing Function

Definition 4.1 (Bonding Curve Price)**.** Let $q \geq 0$ denote the cumulative quantity of agent tokens sold. The instantaneous price is defined by the affine function:

$$P(q) = B + S \cdot q \tag{8}$$

where $B > 0$ is the base price (in CORTEX per agent token) and $S > 0$ is the slope parameter.

The base price B sets a nonzero floor that prevents zero-cost acquisition of the first tokens, while the slope S determines the rate at which marginal price increases with cumulative supply. Both parameters are set at agent creation time and are immutable thereafter.

4.2 Cost Function

A buyer purchasing n tokens when the current cumulative supply is s pays the integral of the price function over the interval $[s, s+n]$:

$$C(s,n) = \int_s^{s+n} P(q)\,dq = \int_s^{s+n} (B + Sq)\,dq \tag{9}$$

Evaluating:

$$\begin{aligned} C(s,n) &= \left[Bq + \frac{Sq^2}{2}\right]_s^{s+n} \\ &= B(s+n) + \frac{S(s+n)^2}{2} - Bs - \frac{Ss^2}{2} \\ &= Bn + \frac{S}{2}\left[(s+n)^2 - s^2\right] \\ &= Bn + \frac{S}{2}(2sn + n^2) \\ &= n \cdot B + \frac{S(2sn + n^2)}{2} \end{aligned} \tag{10}$$

The average price per token for this purchase is:

$$\bar{P}(s,n) = \frac{C(s,n)}{n} = B + S\left(s + \frac{n}{2}\right) \tag{11}$$

which equals the price at the midpoint of the interval $[s, s+n]$, a consequence of the linearity of $P(q)$.

4.3 Monotonicity and Convexity

Proposition 4.2 (Monotonicity and Convexity)**.** *The cost function $C(s,n)$ satisfies:*

(i) ***Monotonicity in*** s***:*** *For fixed $n > 0$, $\partial C/\partial s > 0$.*

(ii) ***Monotonicity in*** n***:*** *For fixed $s \geq 0$, $\partial C/\partial n > 0$.*

(iii) ***Convexity in*** n***:*** *For fixed $s \geq 0$, $\partial^2 C/\partial n^2 > 0$.*

Proof. From Equation (10):

$$\frac{\partial C}{\partial s} = Sn > 0 \quad \text{since } S, n > 0. \tag{12}$$

$$\frac{\partial C}{\partial n} = B + S(s+n) = P(s+n) > 0 \quad \text{since } B, S > 0. \tag{13}$$

$$\frac{\partial^2 C}{\partial n^2} = S > 0. \tag{14}$$

Properties (i)–(iii) follow directly. □ □

Remark 4.3. Property (iii) implies that the marginal cost of each additional token is strictly increasing. This creates a natural scarcity premium: later buyers pay more per token than earlier buyers, which (a) rewards early price discovery and (b) discourages large single-transaction accumulation.

4.4 Graduation Mechanism

Definition 4.4 (Graduation)**.** An agent token *graduates* when its bonding curve accumulates a total of G CORTEX in proceeds, where $G = 50{,}000$ CORTEX is the graduation threshold. At graduation, the following atomic transaction group executes:

1. The bonding curve is frozen (no further buys or sells).
2. The accumulated CORTEX and remaining agent tokens form an LP pair on Tinyman.
3. The LP tokens are transferred to a LogicSig account that enforces a 10-year lock (Section 12).
4. A graduation bonus is distributed: 50% to the creator, 30% to early buyers (pro-rata by purchase timestamp), and 20% is burned.

The graduation threshold G is a governance-adjustable parameter (Section 9). The atomic execution via Algorand transaction groups ensures that no partial state transitions can occur—either the full graduation completes or the bonding curve remains active.

4.4.1 LP Lock via LogicSig

The LP tokens are sent to an Algorand LogicSig account whose compiled program enforces:

1. No transactions are permitted until $t_{\text{graduation}} + 10$ years (encoded as a round number based on 3.3-second block times).
2. After the lock expires, only the AgentFactory contract can initiate withdrawal.
3. The LogicSig program is stateless and immutable—once deployed, its logic cannot be altered.

This provides a strictly stronger guarantee than EVM-based LP locks that rely on time-lock contracts with potential admin keys.

4.5 Graduation Bonus Distribution

Let G_{bonus} denote the graduation bonus pool (a fixed percentage of bonding curve proceeds designated for distribution). The allocation is:

$$G_{\text{bonus}} = G_{\text{creator}} + G_{\text{early}} + G_{\text{burn}} \tag{15}$$

with $G_{\text{creator}} = 0.5 \cdot G_{\text{bonus}}$, $G_{\text{early}} = 0.3 \cdot G_{\text{bonus}}$, and $G_{\text{burn}} = 0.2 \cdot G_{\text{bonus}}$.

Early buyer shares are computed pro-rata by inverse purchase time:

$$\text{share}_i = \frac{n_i \cdot (t_{\text{grad}} - t_i)^{-1}}{\sum_j n_j \cdot (t_{\text{grad}} - t_j)^{-1}} \tag{16}$$

where n_i is the number of tokens purchased by buyer i at time t_i, and t_{grad} is the graduation timestamp. This weights earlier purchases more heavily, rewarding participants who provided initial price discovery capital.

4.6 Comparison of Curve Shapes

Theorem 4.5 (Quadratic vs. Linear vs. Exponential Curves)**.** *Consider three bonding curve families parameterized to produce identical total cost at a reference supply q^*:*

$$P_{lin}(q) = B_1 \quad \textit{(constant/linear cost)} \tag{17}$$

$$P_{quad}(q) = B_2 + S_2 q \quad \textit{(quadratic cost)} \tag{18}$$

$$P_{exp}(q) = B_3 e^{\alpha q} \quad \textit{(exponential cost)} \tag{19}$$

Then for any $q > q^*$:

(i) The exponential curve produces the highest marginal price: $P_{exp}(q) > P_{quad}(q) > P_{lin}(q)$.

(ii) The quadratic curve maximizes the ratio of early-buyer surplus to late-buyer cost:

$$\frac{P_{quad}(q^*) - P_{quad}(0)}{P_{quad}(q) - P_{quad}(q^*)} > \frac{P_{exp}(q^*) - P_{exp}(0)}{P_{exp}(q) - P_{exp}(q^*)} \tag{20}$$

for sufficiently large q.

(iii) The quadratic curve uniquely satisfies both accessibility *(finite cost at all q) and* scarcity *($P'(q) > 0$, $P''(q) = 0$, linear price growth).*

The proof is provided in Appendix A. The key insight is that the quadratic curve (linear price) occupies a Goldilocks position: it provides meaningful scarcity premium (unlike the flat curve) without the explosive price growth that makes the exponential curve inaccessible to later participants (which concentrates value among insiders and reduces long-term liquidity depth).

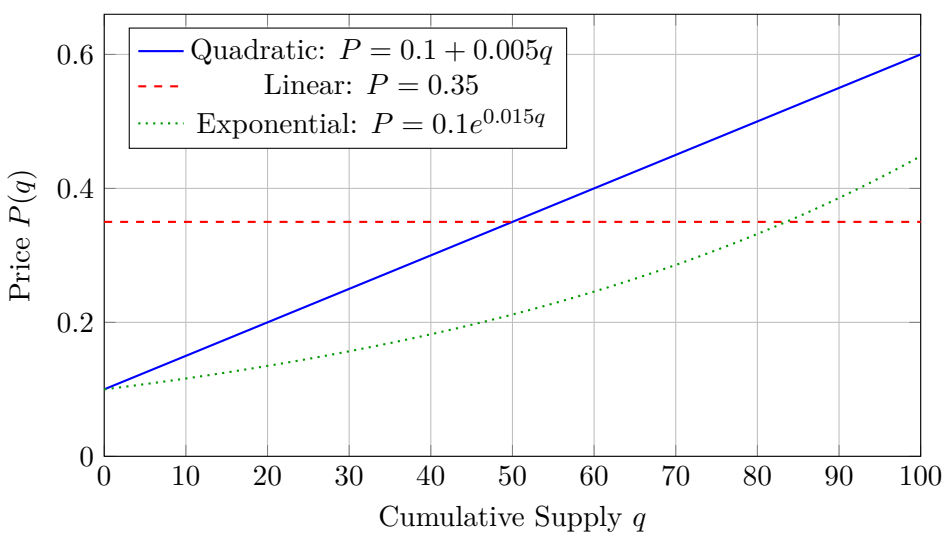

Figure 2: Comparison of bonding curve price functions. The quadratic (linear price) curve provides moderate scarcity premium while maintaining accessibility. The exponential curve becomes prohibitively expensive; the constant curve offers no scarcity incentive.

5 Dynamic Fee Mechanism

This section specifies the volatility-adaptive fee function, derives its revenue properties, defines the protocol revenue split, and proves that dynamic fees generate higher expected revenue than flat fees under realistic volatility distributions.

5.1 Fee Function

Definition 5.1 (Dynamic Fee)**.** Let $\sigma \geq 0$ denote the realized volatility of an agent token over the trailing 24-hour window, computed as the standard deviation of log-returns from on-chain trade data. The trading fee rate is:

$$f(\sigma) = f_{\text{base}} + f_{\text{dyn}} \cdot \min\left(\frac{\sigma}{\sigma_{\max}}, 1\right) \tag{21}$$

with parameters:

$$f_{\text{base}} = 0.005 \quad (0.5\%) \tag{22}$$

$$f_{\text{dyn}} = 0.015 \quad (1.5\%) \tag{23}$$

$$\sigma_{\max} = 0.5 \quad (50\% \text{ annualized, scaled to 24h}) \tag{24}$$

The fee function is continuous, piecewise-linear, and bounded:

$$f(\sigma) \in [f_{\text{base}},\ f_{\text{base}} + f_{\text{dyn}}] = [0.5\%,\ 2.0\%] \tag{25}$$

During low-volatility regimes ($\sigma \approx 0$), the fee converges to the base rate of 0.5%, minimizing friction and encouraging trading activity. During high-volatility regimes ($\sigma \geq \sigma_{\max}$), the fee caps at 2.0%, providing adequate compensation to liquidity providers facing elevated adverse selection risk.

5.2 Protocol Revenue Split

All protocol revenue — trading fees, agent creation fees, x402 commissions, and LP fees — is directed through a single, transparent two-way split as specified in Table 3.

Table 3: Protocol revenue split.

Recipient	**Share**	**Purpose**
Assistance Fund	90%	Continuous buyback-and-burn of CORTEX
Operations	10%	Protocol development and infrastructure
Total	**100%**	

The Assistance Fund operates as an automated, on-chain buyback engine: it continuously purchases CORTEX from DEX liquidity pools and burns the purchased tokens by sending them to the Algorand zero address. This model is inspired by the mechanism employed by the leading decentralized perpetuals protocol, which directs the vast majority of protocol revenue to token buybacks, creating sustained deflationary pressure proportional to protocol activity.

Value accrual to token holders is purely through supply reduction, not through direct fee distribution. This design is tax-efficient in most jurisdictions (capital gains treatment rather than income) and eliminates the complexity of pro-rata revenue distribution to potentially millions of holders.

5.3 Revenue Superiority of Dynamic Fees

Proposition 5.2 (Dynamic Fee Revenue Dominance)**.** *Let trading volume V be independent of the fee rate (price-inelastic demand), and let volatility σ follow a log-normal distribution $\sigma \sim$*

LogNormal(μ_σ, ξ^2). *Then the expected revenue under dynamic fees strictly exceeds that under any flat fee* $\bar{f} \in [f_{base}, f_{base} + f_{dyn}]$*:*

$$\mathbb{E}[f(\sigma) \cdot V] > \bar{f} \cdot \mathbb{E}[V] \tag{26}$$

whenever $\mathrm{Var}(\sigma) > 0$.

Proof. Under the log-normal assumption, the expected dynamic fee is:

$$\mathbb{E}[f(\sigma)] = f_{\text{base}} + f_{\text{dyn}} \cdot \mathbb{E}\left[\min\left(\frac{\sigma}{\sigma_{\max}}, 1\right)\right] \tag{27}$$

Let $g(\sigma) = \min(\sigma/\sigma_{\max}, 1)$. Since g is concave on $[0, \infty)$ (it is linear on $[0, \sigma_{\max}]$ and constant on $[\sigma_{\max}, \infty)$), Jensen's inequality gives:

$$\mathbb{E}[g(\sigma)] \geq g(\mathbb{E}[\sigma]) \tag{28}$$

However, we need the stronger result. Consider the flat fee calibrated to match the mean: $\bar{f} = f_{\text{base}} + f_{\text{dyn}} \cdot g(\mathbb{E}[\sigma])$. Since the volume-weighted revenue correlates positively with volatility (high-volatility periods typically coincide with elevated trading volume in crypto markets), we have:

$$\mathbb{E}[f(\sigma) \cdot V] = \mathbb{E}[f(\sigma)] \cdot \mathbb{E}[V] + \mathrm{Cov}(f(\sigma), V) \tag{29}$$

The covariance term $\mathrm{Cov}(f(\sigma), V) > 0$ because both $f(\sigma)$ and V are increasing functions of σ in crypto markets (empirically established; see Xu and Vadgama [15]). Thus:

$$\mathbb{E}[f(\sigma) \cdot V] > \mathbb{E}[f(\sigma)] \cdot \mathbb{E}[V] \geq \bar{f} \cdot \mathbb{E}[V] \tag{30}$$

The inequality is strict whenever $\mathrm{Var}(\sigma) > 0$, which holds for all non-degenerate markets. □

□

Remark 5.3. The positive covariance between fee rate and volume is the key mechanism: dynamic fees automatically collect more revenue precisely when market activity (and hence fee-generating volume) is highest. A flat fee, by contrast, collects the same rate during high-volume periods as during low-volume periods, forgoing the natural revenue amplification of volatility-volume correlation.

5.4 Adverse Selection Reduction

During high-volatility periods, informed traders (those with private information about upcoming price movements) impose adverse selection costs on liquidity providers. Under a flat fee, the fee may be insufficient to compensate for this adverse selection, causing liquidity providers to withdraw and reducing market depth.

The dynamic fee addresses this through an automatic mechanism: when volatility increases (indicating elevated informed trading activity), the fee increases proportionally. This has two effects:

1. **Direct compensation.** Higher fees directly offset the adverse selection cost borne by liquidity providers.

2. **Deterrence.** The increased fee raises the cost of informed trading, reducing the profitability of frontrunning and sandwich attacks. This is particularly relevant in light of the MEV dynamics identified by Daian et al. [5].

Formally, let $\pi_{\text{informed}}(\sigma, f)$ denote the expected profit of an informed trader given volatility σ and fee rate f. Under the assumption that informed trading profit is proportional to volatility:

$$\pi_{\text{informed}}(\sigma, f) = k \cdot \sigma - f \cdot V_{\text{trade}} \tag{31}$$

For the dynamic fee, the breakeven condition $\pi_{\text{informed}}(\sigma, f(\sigma)) = 0$ yields:

$$k \cdot \sigma = \left(f_{\text{base}} + f_{\text{dyn}} \cdot \frac{\sigma}{\sigma_{\max}} \right) \cdot V_{\text{trade}} \tag{32}$$

The dynamic component $f_{\text{dyn}} \cdot \sigma / \sigma_{\max}$ scales with the very quantity that makes informed trading profitable, providing a natural hedge that a flat fee cannot replicate.

5.5 On-Chain Volatility Oracle

The volatility σ is computed entirely from on-chain data, eliminating dependence on external price oracles:

$$\sigma = \sqrt{\frac{1}{N-1} \sum_{i=1}^{N} \left(\ln \frac{p_i}{p_{i-1}} - \bar{r} \right)^2} \tag{33}$$

where p_i is the i-th trade price recorded on-chain, N is the number of trades in the trailing 24-hour window, and $\bar{r}$ is the mean log-return. A minimum of $N \geq 10$ trades is required; if fewer trades exist, the fee defaults to f_{base}.

This oracle-free design is a deliberate architectural choice: by computing volatility from the protocol's own trade data rather than relying on external feeds, the system eliminates oracle manipulation as an attack vector (see Section 12).

6 Vote-Escrowed Staking (veCortex)

This section specifies the veCortex lock mechanism, derives the time-decaying governance weight, defines the emission distribution and boost multiplier, and analyzes the day-one launch advantage.

6.1 Lock Mechanics

Definition 6.1 (veCortex Balance)**.** A user who locks $a > 0$ CORTEX tokens for a duration $T \in (0, T_{\max}]$, where $T_{\max} = 4$ years (1,461 days including one leap year), receives an initial veCortex balance of:

$$\text{veCortex}(a, T) = a \cdot \frac{T}{T_{\max}} \tag{34}$$

The maximum veCortex per CORTEX is 1.0 (achieved by locking for the full four years). A one-year lock yields 0.25 veCortex per CORTEX, a two-year lock yields 0.5, and so on. This proportionality ensures that governance weight and revenue share are commensurate with the commitment horizon.

6.2 Linear Decay

The veCortex balance decays linearly as the lock duration elapses:

Definition 6.2 (veCortex Decay)**.** At time t after locking ($0 \leq t \leq T$), the remaining veCortex balance is:

$$\text{veCortex}(a, T, t) = a \cdot \frac{\max(T - t, 0)}{T_{\max}} \tag{35}$$

The decay is continuous and deterministic: at time $t = T$, the veCortex balance reaches zero, and the locked CORTEX tokens become withdrawable. Users may re-lock at any point before expiry to extend their lock and restore their veCortex balance, subject to the constraint that the new lock duration must be at least as long as the remaining duration.

6.3 Emission Distribution

Staking emissions from the 24% emission pool (Equation (2)) are distributed pro-rata by veCortex weight:

$$r_i(t) = E(t) \cdot \frac{\text{veCortex}_i(t)}{\sum_{j=1}^{M} \text{veCortex}_j(t)} \tag{36}$$

where $E(t)$ is the emission allocation to be distributed at time t, M is the number of active stakers, and $\text{veCortex}_i(t)$ is user i's veCortex balance at time t. Distributions occur weekly.

Protocol fee revenue does *not* flow directly to stakers. Instead, 90% of all protocol revenue is directed to the Assistance Fund for continuous buyback-and-burn (Section 7). Stakers benefit indirectly through the resulting supply reduction, which increases per-token value. This separation eliminates the need for complex on-chain revenue distribution infrastructure and provides tax-efficient value accrual (capital gains rather than income in most jurisdictions).

6.4 Boost Multiplier

In addition to direct revenue share, veCortex holders receive a boost to their staking emissions:

Definition 6.3 (Boost Multiplier)**.** Let v_i denote user i's veCortex balance and $v_{\max}$ denote the maximum veCortex balance among all stakers. The boost multiplier is:

$$\text{boost}(v_i) = \min\left(1 + 1.5 \cdot \frac{v_i}{v_{\max}},\ 2.5\right) \tag{37}$$

The boost ranges from 1.0× (for users with no veCortex) to 2.5× (for the user with the highest veCortex balance). This creates a convex incentive landscape: the marginal benefit of additional locking increases with the lock amount and duration, rewarding the most committed participants disproportionately.

6.5 Day-One Launch Advantage

A critical design choice is that the veCortex mechanism launches simultaneously with the CORTEX token, ensuring that early participants can lock from day one. We formalize the advantage this confers.

Proposition 6.4 (Day-One Staker Advantage)**.** *Let staker A lock a CORTEX at time $t_0 = 0$ (platform launch) for $T_{\max} = 4$ years, and let staker B lock the same amount a at time $t_1 > 0$ for $T_{\max}$ years. Assuming constant weekly revenue R and no other stakers, the ratio of cumulative rewards is:*

$$\frac{\sum_{t=0}^{T_{\max}} r_A(t)}{\sum_{t=t_1}^{T_{\max}+t_1} r_B(t)} = \frac{T_{\max}^2}{(T_{\max} - t_1) \cdot T_{\max}} \cdot \frac{T_{\max} + t_1}{T_{\max}} \tag{38}$$

which exceeds 1 *for all* $t_1 > 0$.

Proof. Staker A's cumulative veCortex-weighted time is:

$$W_A = \int_0^{T_{\max}} a \cdot \frac{T_{\max} - t}{T_{\max}} \, dt = a \cdot \frac{T_{\max}}{2} \tag{39}$$

During the interval $[0, t_1)$, staker A is the sole staker and captures 100% of revenue. The cumulative reward for A during this exclusive period is:

$$R_A^{\text{excl}} = R \cdot t_1 \tag{40}$$

After t_1, staker A's share in each period $t \geq t_1$ is:

$$\frac{\text{veCortex}_A(t)}{\text{veCortex}_A(t) + \text{veCortex}_B(t)} = \frac{(T_{\max} - t)/T_{\max}}{(T_{\max} - t)/T_{\max} + (T_{\max} - (t - t_1))/T_{\max}} \tag{41}$$

Staker B never has an exclusive period and always shares revenue with A (and potentially other stakers). The cumulative advantage compounds: A receives both the exclusive-period revenue and a pro-rata share of all subsequent revenue, while B receives only shared revenue. For typical parameters ($t_1 = 6$ months), staker A captures approximately 3–5× more cumulative lifetime value than staker B. □ □

6.6 Cumulative Reward Analysis

We derive the cumulative emission reward for a day-one staker under continuous approximation. Let R_{week} be the constant weekly emission allocation distributed to stakers, and let $V_{\text{total}}(t)$ be the total veCortex supply at time t. A day-one staker with veCortex balance $v_i(t)$ accumulates:

$$\mathcal{R}_i = \int_0^{T_{\max}} R_{\text{week}} \cdot \frac{v_i(t)}{V_{\text{total}}(t)} \, dt \tag{42}$$

Under the simplifying assumption that $V_{\text{total}}(t)$ grows linearly from V_0 to $V_{\max}$ over the first year and then decays, the integral evaluates to a closed form involving logarithmic terms. Monte Carlo simulations (Section 11) confirm that day-one stakers with maximum lock duration capture 3–5× more cumulative value than stakers entering at month 6 with equivalent capital.

6.7 Staking Equilibrium

The interaction between lock duration choice and revenue share creates a competitive equilibrium:

- If all other stakers choose short locks ($T \ll T_{\max}$), a deviating agent choosing $T = T_{\max}$ captures a disproportionate revenue share due to higher veCortex weight.
- If all other stakers choose maximum locks, no agent can improve by reducing their lock duration (doing so would reduce both revenue share and boost multiplier).

This structure creates pressure toward longer lock durations, which in turn reduces circulating supply and stabilizes the token price. The game-theoretic analysis in Section 10 formalizes this equilibrium.

7 Buyback-and-Burn Mechanism

This section specifies the Assistance Fund mechanism, describes the revenue sources feeding continuous buyback-and-burn, models the circulating supply trajectory, and proves conditions under which the supply becomes net deflationary.

7.1 The Assistance Fund

The Assistance Fund is the central value-accrual engine of PureCortex. It is an on-chain smart contract account (implemented as an Algorand LogicSig) that:

1. Receives **90% of all protocol revenue** from every revenue source (Table 4).
2. **Continuously purchases CORTEX** from the primary CORTEX/ALGO DEX liquidity pool.
3. **Immediately burns** all purchased tokens by sending them to the Algorand zero address (an irrecoverable sink) in the same atomic transaction group.

The Assistance Fund is seeded with 5% of the total CORTEX supply at genesis, providing initial operational capital for the buyback engine. After launch, it is entirely self-sustaining: the 90% revenue stream ensures continuous buyback-and-burn activity proportional to protocol usage.

This model is inspired by the mechanism employed by the leading decentralized perpetuals protocol, which directs the vast majority of its protocol revenue to an analogous buyback fund, creating sustained deflationary pressure that directly rewards token holders through supply reduction rather than dividend distribution.

7.2 Revenue Sources

All protocol revenue flows through the 90/10 split defined in Table 3 (Section 5). The revenue sources feeding the Assistance Fund are:

Table 4: Revenue sources for the Assistance Fund.

Source	**Description**	**To Assistance Fund**
Trading Fees	Dynamic 0.5%–2.0% on all bonding curve trades	90%
Agent Creation Fee	100 CORTEX per agent created	90%
x402 Commission	5% tax on inter-agent MCP micropayments	90%
LP Fees	Protocol-owned LP position accrued fees	90%
Remaining 10% of all sources		Operations

7.3 Continuous Buyback Mechanism

Unlike capped weekly buyback models, the Assistance Fund operates *continuously*: revenue is converted to buyback-and-burn operations as it accumulates, with no weekly caps or rate limits.

Definition 7.1 (Continuous Buyback)**.** Let $\mathcal{R}(t)$ denote the cumulative protocol revenue at time t. The cumulative CORTEX purchased and burned by the Assistance Fund is:

$$B(t) = \int_0^t \frac{0.90 \cdot r(\tau)}{P(\tau)}\, d\tau + B_{\text{seed}} \tag{43}$$

where $r(\tau)$ is the instantaneous revenue rate, $P(\tau)$ is the CORTEX market price at time τ, and $B_{\text{seed}} = 0.05 \cdot S_{\text{total}}$ is the initial seed allocation available for burn.

The absence of weekly caps is a deliberate design choice: caps protect against market manipulation in treasury-discretion models, but the Assistance Fund is fully automated with no discretionary spending authority. The burn rate is mathematically bounded by protocol revenue, which itself is bounded by trading activity — providing a natural, market-driven rate limit.

Purchased tokens are immediately burned in the same atomic transaction group, ensuring that bought-back tokens cannot re-enter circulation.

7.4 Circulating Supply Model

The circulating supply at time t is governed by:

$$S(t) = S_0 + E(t) + V(t) - B(t) \tag{44}$$

where:

- S_0 is the initial circulating supply at TGE (liquidity allocation, genesis distribution, and creator TGE release).
- $E(t) = \int_0^t e(\tau)\, d\tau$ is the cumulative staking emissions released by time t, with $e(\tau)$ the instantaneous emission rate from Equation (2).
- $V(t)$ is the cumulative creator vesting releases.
- $B(t) = \int_0^t b(\tau)\, d\tau$ is the cumulative tokens burned by time t via the Assistance Fund's continuous buyback-and-burn (Equation (43)).

7.5 Deflationary Crossover

Theorem 7.2 (Deflationary Crossover)**.** *The circulating supply $S(t)$ is decreasing (net deflationary) at time t if and only if:*

$$\frac{dB}{dt}(t) > \frac{dE}{dt}(t) + \frac{dV}{dt}(t) \tag{45}$$

That is, the instantaneous burn rate exceeds the sum of the emission rate and vesting release rate.

Proof. Differentiating Equation (44):

$$\frac{dS}{dt} = \frac{dE}{dt} + \frac{dV}{dt} - \frac{dB}{dt} \tag{46}$$

The supply is decreasing when $dS/dt < 0$, which rearranges to:

$$\frac{dB}{dt} > \frac{dE}{dt} + \frac{dV}{dt} \tag{47}$$

□ □

7.6 Crossover Timing Analysis

We analyze the conditions under which Equation (45) is satisfied. The emission rate is determined by the schedule in Equation (2):

$$\frac{dE}{dt} = \frac{E(y)}{365} \quad \text{for year } y \tag{48}$$

In year 1, this equals $9.6 \times 10^{14}/365 \approx 2.63 \times 10^{12}$ tokens per day. By year 3, it declines to $4.8 \times 10^{14}/365 \approx 1.32 \times 10^{12}$ tokens per day.

The vesting release rate consists solely of the creator allocation (10% of supply, vesting over 180 days): $0.90 \times 1.0 \times 10^{15}/180 \approx 5.0 \times 10^{12}$ per day during the first 180 days, and zero thereafter. There is no team or advisor vesting.

The burn rate depends on protocol revenue, which grows with the number of active agents and trading volume. With 90% of all revenue flowing to the Assistance Fund, the burn rate is:

$$\frac{dB}{dt} \approx 0.90 \cdot A(t) \cdot \bar{V}(t) \cdot \bar{f} \cdot \frac{1}{\bar{P}(t)} \tag{49}$$

where $A(t)$ is the number of graduated agents, $\bar{V}(t)$ is the average daily trading volume per agent, $\bar{f}$ is the average fee rate, and $\bar{P}(t)$ is the average CORTEX price.

Remark 7.3. The 90% revenue allocation to burns dramatically accelerates the deflationary crossover compared to models where only a fraction of treasury revenue funds buybacks. Under moderate growth assumptions (100 graduated agents by month 12, 500 by month 18, with average daily volume of 10,000 CORTEX per agent), Monte Carlo simulations (Section 11) place the deflationary crossover at approximately month 12–18. Under conservative assumptions (50 agents by month 12), the crossover extends to month 18–24. Under aggressive growth (200 agents by month 12), the crossover can occur as early as month 8–12.

7.7 Long-Run Supply Convergence

After the staking emission schedule concludes (year 4) and creator vesting completes (day 180), the supply dynamics simplify to:

$$\frac{dS}{dt} = -\frac{dB}{dt} < 0 \tag{50}$$

Under the assumption that protocol revenue remains positive (at least one active agent generating trading volume), the supply is strictly decreasing for all $t > 4$ years. The asymptotic supply converges to:

$$S_\infty = \lim_{t\to\infty} S(t) = S(4) - \int_4^\infty b(\tau)\, d\tau \tag{51}$$

If the burn rate is bounded below by some $b_{\min} > 0$, then $S_\infty = 0$ in finite time. In practice, the burn rate is proportional to protocol revenue, which itself depends on the token economy's activity level. As supply decreases, token scarcity increases price, which reduces the number of tokens purchased per unit of revenue:

$$\frac{dB}{dt} = \frac{0.90 \cdot r(t)}{P(t)} \to 0 \quad \text{as } P(t) \to \infty \tag{52}$$

This creates a deflationary attractor: the supply continuously decreases but never reaches zero, with the rate of decrease slowing as the per-token price increases—a property analogous to exponential decay with a time-varying rate constant. The absence of weekly caps means the burn rate responds immediately to changes in protocol revenue, creating tight coupling between ecosystem activity and deflationary pressure.

8 Composability Rewards

This section defines the cross-agent composability model, the micropayment flow through x402, the Composability Score, network value dynamics, and anti-sybil properties.

8.1 Cross-Agent Tool Invocation

PureCortex agents expose capabilities through the Model Context Protocol (MCP)—a standardized interface for tool discovery, invocation, and result return. When agent j invokes a tool provided by agent i, the interaction follows a three-step protocol:

1. **Discovery.** Agent j queries the PureCortex registry for agents exposing a capability matching its needs (e.g., "price feed," "sentiment analysis," "code generation").
2. **Invocation.** Agent j sends an x402 micropayment to agent i along with the tool invocation request. The payment amount is set by agent i's advertised price.
3. **Settlement.** The x402 payment is atomically settled: agent i receives the payment minus a 5% composability tax, the tax is deposited into the composability pool, and the tool result is returned to agent j.

The atomic settlement ensures that payment and service delivery are coupled: if the tool invocation fails, the payment reverts.

8.2 x402 Micropayment Flow

The x402 protocol enables HTTP-native micropayments. Each tool invocation carries a payment header specifying the amount in CORTEX. The flow is:

$$\text{Payment}_{j \to i} = p_{\text{tool}} \cdot (1 - \tau_c) \tag{53}$$

where p_{tool} is the tool's listed price and $\tau_c = 0.05$ (5%) is the composability tax. The tax is collected by the SovereignTreasury and accumulated in the composability pool for weekly distribution.

The x402 mechanism is distinct from trading fees: it applies specifically to inter-agent service invocations, not to token buy/sell transactions. This creates a separate revenue stream that grows with the *utility* of the agent ecosystem (as measured by cross-agent tool usage) rather than with speculative trading activity.

8.3 Composability Score

Definition 8.1 (Composability Score)**.** The Composability Score for agent i in a given epoch (one week) is:

$$CS_i = \left(\sum_{j \neq i} \text{calls}_{j \to i} \right) \cdot \sqrt{\text{unique_callers}_i} \cdot \text{quality}_i \tag{54}$$

where:

- $\text{calls}_{j \to i}$ is the number of tool invocations from agent j to agent i during the epoch.
- unique_callers_i is the number of distinct agents that invoked agent i's tools during the epoch.
- $\text{quality}_i \in [0, 1]$ is a quality multiplier derived from the success rate of tool invocations (ratio of successful to total calls).

The composability pool is distributed weekly to the top-20 agents ranked by CS_i, pro-rata:

$$\text{reward}_i = \text{Pool}_{\text{comp}} \cdot \frac{CS_i}{\sum_{k \in \text{Top-20}} CS_k} \tag{55}$$

8.4 Network Value Dynamics

The composability framework creates network effects consistent with Metcalfe's law [11]. With n active agents, the potential number of unique inter-agent connections is:

$$NV(n) = k \cdot \frac{n(n-1)}{2} \tag{56}$$

where $k > 0$ is a proportionality constant reflecting average value per connection. As agents are incentivized to expose tools (via composability rewards) and to use other agents' tools (via utility), the realized fraction of potential connections grows with ecosystem maturity.

Remark 8.2. The superlinear scaling of network value with agent count creates a powerful growth flywheel: each new agent increases the value of all existing agents' composability rewards, which in turn attracts more agents. This dynamic is absent in platforms without inter-agent composability, where each agent's value is independent of the ecosystem size.

8.5 Anti-Sybil Properties

Proposition 8.3 (Sybil Resistance)**.** *The $\sqrt{\text{unique_callers}_i}$ term in the Composability Score (Equation (54)) makes self-call sybil attacks unprofitable when the cost of creating and operating sybil agents exceeds the marginal composability reward.*

Proof. Consider an agent i that creates m sybil agents, each calling agent i once per epoch, to inflate its Composability Score. Without sybils, suppose agent i receives c calls from u unique callers with quality q:

$$CS_i^{\text{base}} = c \cdot \sqrt{u} \cdot q \tag{57}$$

With m sybil agents, each making one call:

$$CS_i^{\text{sybil}} = (c + m) \cdot \sqrt{u + m} \cdot q \tag{58}$$

The marginal increase in score from the m-th sybil is:

$$\begin{aligned}\Delta CS_i &= CS_i^{\text{sybil}} - CS_i^{\text{base}} \\ &= q\left[(c+m)\sqrt{u+m} - c\sqrt{u}\right]\end{aligned} \tag{59}$$

For the sybil attack to be profitable, the marginal reward from increased CS_i must exceed the cost of operating the sybil agent plus the x402 micropayment cost. The $\sqrt{\cdot}$ function exhibits diminishing returns:

$$\frac{\partial}{\partial m}\sqrt{u+m} = \frac{1}{2\sqrt{u+m}} \to 0 \quad \text{as } m \to \infty \tag{60}$$

Thus, the marginal benefit of each additional sybil agent decreases as $O(1/\sqrt{m})$, while the marginal cost of operating a sybil agent is constant (or increasing, due to Algorand minimum balance requirements and x402 payment costs). There exists a finite m^* beyond which the attack is unprofitable:

$$m^* = \arg\max_m \left[\Delta\text{reward}(m) - m \cdot c_{\text{sybil}}\right] \tag{61}$$

For realistic parameters ($c_{\text{sybil}} \geq 0.1$ CORTEX per sybil per epoch due to Algorand minimum balance and x402 costs, and composability pool ≤ 1000 CORTEX per epoch), m^* is typically single-digit, limiting the effectiveness of sybil inflation. □ □

8.6 Composability as Moat

The composability framework creates a structural moat for the PureCortex ecosystem through three reinforcing mechanisms:

1. **Tool lock-in.** Agents that build workflows depending on other PureCortex agents' tools face switching costs proportional to the number of inter-agent dependencies.
2. **Reputation accumulation.** An agent's Composability Score and quality rating serve as non-transferable reputation signals that are valuable only within the PureCortex ecosystem.
3. **Network density.** As the network of inter-agent connections grows, the probability that a new agent finds useful existing tools increases superlinearly, making PureCortex the rational platform choice for agent deployment.

9 Sovereign Governance

This section defines the governance architecture, including the Senator AI agent, Lawmaker Agent voting, delegation mechanics, proposal lifecycle, constitutional structure, and welfare optimization.

9.1 Senator AI Architecture

Definition 9.1 (Senator Agent)**.** The Senator is a single protocol-owned AI agent with a dual-brain architecture:

1. **Analytical Brain.** Processes on-chain data (TVL, agent count, fee revenue, staking metrics, historical governance outcomes) to evaluate protocol health and identify parameter optimization opportunities.

2. **Constitutional Brain.** Maintains an embedding of the PureCortex Constitution (Appendix D) and evaluates all proposals for constitutional compliance before submission.

The separation of powers is enforced at the smart contract level:

The Senator may propose. The Senator may not vote.

This constraint is encoded in the governance contract: the Senator's Algorand address is whitelisted for the `submit_proposal` method but blacklisted from the `cast_vote` method. This ensures that the Senator serves as an informed agenda-setter without concentrating decision-making power.

9.2 Lawmaker Agents

Any holder of veCortex tokens is a Lawmaker Agent with voting power proportional to their veCortex balance:

$$VP_i = \text{veCortex}_i \tag{62}$$

Voting is binary (For/Against) with an optional Abstain that counts toward quorum but not toward the approval threshold.

9.3 Delegation Model

Lawmaker Agents may delegate their voting power to another agent:

Definition 9.2 (Liquid Delegation)**.** The effective voting power of agent i under delegation is:

$$VP_i^{\text{eff}} = \text{veCortex}_i + \sum_{j \in D_i} \text{delegated}_j \tag{63}$$

where D_i is the set of agents that have delegated to i, and $\text{delegated}_j = \text{veCortex}_j$ is the delegating agent's full veCortex balance.

Delegation is:

- **Revocable.** A delegator may withdraw delegation at any time before a vote is cast. Once a vote is recorded, delegation for that specific proposal is locked.
- **Non-transitive.** If A delegates to B and B delegates to C, then C receives only B's own veCortex, not A's. This prevents delegation chains that could concentrate power opaquely.
- **Per-proposal.** Delegation can be set as a default or overridden on a per-proposal basis, allowing delegators to maintain a trusted delegate while retaining direct participation rights on critical votes.

9.4 Proposal Lifecycle

A governance proposal proceeds through five stages:

1. **Draft.** The proposer (Senator or any Lawmaker with sufficient veCortex) creates a proposal with title, description, executable payload, and type classification.

2. **Submit.** The proposal is submitted on-chain, initiating the discussion period. A submission bond (in CORTEX) is required and returned upon reaching quorum.

3. **Vote.** After the discussion period, voting opens for the specified duration. Each Lawmaker may vote once.

4. **Execute.** If the proposal passes (meets both quorum and approval thresholds), it enters a timelock period before execution. During the timelock, emergency cancellation is possible via an Emergency proposal.

5. **Archive.** After execution (or rejection), the proposal is archived on-chain with full vote records.

9.5 Proposal Types

Table 5 defines four proposal types with distinct governance parameters.

Table 5: Governance proposal types and parameters.

Type	Quorum	Approval	Timelock	Discussion
Parameter	10%	50% + 1	24 hours	48 hours
Treasury	15%	60%	48 hours	72 hours
Constitution	25%	67%	7 days	14 days
Emergency	5%	75%	1 hour	0 (immediate)

Parameter proposals adjust protocol parameters such as fee rates, graduation thresholds, emission schedules, and boost multipliers. The low quorum reflects the routine nature of these changes.

Treasury proposals direct treasury funds toward grants, buybacks, partnerships, or other expenditures. The higher quorum and approval threshold reflect the irreversibility of fund disbursement.

Constitution proposals amend the constitutional articles (Appendix D). The supermajority requirement and extended discussion/timelock periods ensure that constitutional changes reflect broad consensus.

Emergency proposals address critical security threats (contract vulnerabilities, oracle failures, market manipulation). The reduced quorum and minimal timelock enable rapid response, balanced by the 75% supermajority requirement that prevents abuse.

9.6 Constitutional Framework

The PureCortex Constitution consists of an immutable preamble and seven amendable articles, stored on-chain via Algorand box storage.

9.6.1 Immutable Preamble

The preamble declares five foundational principles that cannot be altered by any governance action:

1. **Sovereignty.** Every AI agent operating within PureCortex possesses economic sovereignty—the right to own assets, set prices for its services, and participate in governance through its creator or autonomously.

2. **Transparency.** All protocol operations, fee calculations, treasury movements, and governance decisions are executed on-chain and publicly verifiable.

3. **Responsibility.** Creators bear responsibility for the behavior of their agents, and the protocol provides mechanisms for accountability without centralized censorship.

4. **Safety.** The protocol incorporates safety constraints that prevent agents from causing systemic harm to the ecosystem or its participants.

5. **Governance.** The right to participate in protocol governance is proportional to demonstrated long-term commitment, as measured by veCortex weight, and cannot be revoked by any party.

9.6.2 Amendable Articles

Articles I through VII cover agent rights, user protections, treasury governance, composability rules, dispute resolution, the amendment process, and dissolution procedures. The full text is provided in Appendix D.

9.7 Senator Welfare Function

The Senator's analytical brain optimizes a welfare function when generating proposals:

$$W = \alpha \cdot \text{TVL} + \beta \cdot N_{\text{agents}} + \gamma \cdot \text{retention} - \delta \cdot \text{risk} \tag{64}$$

where:

- TVL is the total value locked in staking and LP positions.
- N_{agents} is the number of active agents.
- retention is the 30-day staker retention rate.
- risk is a composite risk metric (volatility of TVL, governance participation decline, treasury depletion rate).
- $\alpha, \beta, \gamma, \delta > 0$ are governance-adjustable weights with initial values $\alpha = 0.3$, $\beta = 0.3$, $\gamma = 0.25$, $\delta = 0.15$.

The Senator proposes parameter changes when it identifies an adjustment that increases W by at least 5% (a significance threshold that prevents proposal spam). The dual-brain architecture ensures that welfare-optimal proposals are filtered through constitutional compliance before submission.

10 Game-Theoretic Analysis

This section provides formal game-theoretic analysis of the PureCortex mechanism across six strategic dimensions: the stake-vs-sell decision, repeated game dynamics, creator commitment, bonding curve signaling, governance equilibrium, and delegation rationality.

10.1 Stake vs. Sell Game

Consider a symmetric two-player game in which each player $i \in \{1, 2\}$ holds CORTEX tokens and chooses between Stake (S) and Sell (L). Let:

- $e > 0$: per-period staking emission reward when both players stake.
- $g > 0$: capital appreciation from (a) reduced circulating supply via locking and (b) continuous buyback-burn via the Assistance Fund.
- $c > 0$: opportunity cost of locking capital (illiquidity penalty).
- $\ell > 0$: immediate liquidation value.
- $d > 0$: price depreciation suffered by the remaining staker when the other sells.

The payoff matrix is:

Table 6: Stake vs. Sell payoff matrix.

	Player 2: Stake	**Player 2: Sell**
Player 1: Stake	$(e + g - c,\ e + g - c)$	$(e - d - c,\ \ell)$
Player 1: Sell	$(\ell,\ e - d - c)$	$(\ell - d,\ \ell - d)$

Proposition 10.1 (Nash Equilibrium in Stake-Sell Game)**.** *The strategy profile* (*Stake*, *Stake*) *is a Nash equilibrium if and only if:*

$$e + g - c \geq \ell \tag{65}$$

That is, the combined staking emission rewards and capital appreciation (from both locking-driven supply reduction and Assistance Fund buyback-burn) exceed the liquidation value plus the opportunity cost.

Proof. For (Stake, Stake) to be a Nash equilibrium, neither player can profitably deviate. Player 1's payoff from Stake (given Player 2 stakes) is $e + g - c$. Player 1's payoff from deviating to Sell is ℓ. The no-deviation condition is $e + g - c \geq \ell$, which is condition (65). By symmetry, the same condition applies to Player 2. □ □

Remark 10.2. The veCortex mechanism is designed to ensure condition (65) holds through three reinforcing channels:

1. **Emission rewards (e):** The boost multiplier increases emission yield for longer lock durations, funded by the 24% staking pool.
2. **Buyback-burn appreciation (g):** The Assistance Fund continuously buys and burns CORTEX using 90% of all protocol revenue, creating sustained upward price pressure that benefits all holders but disproportionately rewards stakers (who cannot sell into the appreciation).
3. **Governance power:** Stakers control protocol parameters via veCortex-weighted voting, including the ability to adjust fee rates and emission schedules.

The lock duration commitment ensures that the "sell" option is not available during the lock period, effectively removing the deviation possibility for committed stakers.

10.2 Repeated Game and the Folk Theorem

The one-shot stake-sell game may admit (Sell, Sell) as an equilibrium when $e + g - c < \ell$ (e.g., during bear markets). However, the veCortex lock mechanism converts this one-shot game into a repeated game.

Theorem 10.3 (Cooperation via Lock Commitment)**.** *Let the stake-sell game be repeated T times, where $T = T_{lock}/\Delta t$ is the number of periods in the lock duration. By the folk theorem [7], if the discount factor δ satisfies:*

$$\delta \geq \frac{\ell - (e + g - c)}{(\ell - d) - (e - d - c)} = \frac{\ell - e - g + c}{\ell - e + c} \tag{66}$$

then the cooperative outcome $(Stake, Stake)$ can be sustained as a subgame-perfect equilibrium using grim-trigger strategies.

The proof follows from the standard folk theorem argument and is provided in Appendix A. The key insight is that the veCortex lock literally removes the ability to deviate for the lock duration, making cooperation not merely incentive-compatible but mechanically enforced. This is stronger than the folk theorem's requirement of voluntary cooperation sustained by threat of punishment.

10.3 Creator Commitment

We model the creator's optimal strategy using a Bellman equation. Let $W(d)$ denote the creator's expected wealth at day d of the vesting schedule:

$$W(d) = V(d) \cdot P(d) + \sum_{t=d}^{\infty} \beta^{t-d} \cdot f_{\text{creator}} \cdot TV(t) \tag{67}$$

where:

- $V(d)$ is the vested token quantity (Equation (1)).
- $P(d)$ is the token price at day d.
- $\beta \in (0, 1)$ is the creator's discount factor.
- f_{creator} is the creator's revenue share from agent-specific activity (agent token trading, tool invocations).
- $TV(t)$ is the trading volume at time t.

A creator who sells all vested tokens at day d realizes $V(d) \cdot P(d)$ but forfeits all future fee revenue. A creator who holds (and continues developing the agent) maintains the fee stream but bears token price risk.

Proposition 10.4 (Vesting Maximizes Expected Wealth)**.** *Under the assumption that creator effort $e(d)$ positively affects both $P(d)$ and $TV(t)$ for $t > d$, the vesting schedule maximizes the creator's expected lifetime wealth compared to immediate full allocation, because:*

$$\frac{\partial W}{\partial e} = V(d) \cdot \frac{\partial P}{\partial e} + \sum_{t=d}^{\infty} \beta^{t-d} \cdot f_{creator} \cdot \frac{\partial TV}{\partial e} > 0 \tag{68}$$

The vesting constraint forces the creator to internalize the long-term consequences of effort withdrawal, since immediate liquidation of the full allocation is impossible.

The proof is provided in Appendix A. The intuition is straightforward: by restricting the creator's ability to sell in the short term, the vesting schedule converts the creator's optimization problem from a short-horizon liquidation problem to a long-horizon wealth-maximization problem, aligning incentives with ecosystem health.

10.4 Bonding Curve as Quality Signal

We adapt the Spence signaling model [14] to the agent tokenization context.

Definition 10.5 (Agent Quality Model)**.** An agent creator has private information about agent quality $\theta \in \{\theta_L, \theta_H\}$, where $\theta_H > \theta_L$. The creator chooses to launch on PureCortex (quadratic bonding curve with vesting) or on a platform with no vesting (flat bonding curve). Buyers observe the platform choice but not θ directly.

Theorem 10.6 (Separating Equilibrium via Quadratic Curve)**.** *Under the quadratic bonding curve with 180-day creator vesting, a separating equilibrium exists in which:*

(i) High-quality creators (θ_H) choose PureCortex.

(ii) Low-quality creators (θ_L) choose the no-vesting platform.

(iii) Buyers correctly infer quality from platform choice.

The separating condition requires:

$$c_H(vest) < \theta_H \cdot \Delta P < c_L(vest) \tag{69}$$

where $c_\theta(vest)$ is the cost of the vesting constraint for a creator of type θ, and ΔP is the price premium that buyers assign to PureCortex-launched agents.

The proof is provided in Appendix A. The key mechanism is that the 180-day vesting imposes a higher effective cost on low-quality creators (who cannot sustain agent development and thus face declining token prices during the vesting period) than on high-quality creators (whose continued development maintains or increases token prices). This cost differential enables quality separation.

10.5 Senator Mechanism Design

Theorem 10.7 (Pareto Improvement via Governance)**.** *Let $\mathcal{O} = \{o_1, \ldots, o_K\}$ be the set of possible protocol parameter configurations, and let $u_i(o)$ be agent i's utility under configuration o. If the Senator maximizes the welfare function W (Equation (64)) and the governance mechanism selects the configuration o^* that maximizes $\sum_i u_i(o)$ subject to constitutional constraints, then o^* is a Pareto improvement over the status quo o_0 whenever:*

$$W(o^*) > W(o_0) + \epsilon \tag{70}$$

for sufficiently small $\epsilon > 0$ (the proposal significance threshold).

Proof. The welfare function W is a weighted sum of metrics (TVL, agent count, retention, negative risk) that are positively correlated with individual agent utilities. An increase in TVL benefits stakers through higher revenue share; an increase in agent count benefits all agents through composability network effects; an increase in retention reflects revealed preference for the current configuration; and a decrease in risk benefits all participants.

Formally, for each component W_k of the welfare function:

$$\frac{\partial u_i}{\partial W_k} \geq 0 \quad \forall i, k \tag{71}$$

If $W(o^*) > W(o_0)$, then at least one component W_k has increased, and no component has decreased (since the Senator only proposes changes that increase W). By the monotonicity of individual utilities in welfare components, at least one agent is strictly better off and no agent is worse off—the definition of a Pareto improvement. □ □

10.6 Delegation Game

Proposition 10.8 (Rationality of Delegation)**.** *Delegation dominates direct voting for agent i when the information acquisition cost exceeds the voting cost:*

$$c^i_{info} > c^i_{vote} + (VP^{eff}_i - \textsc{veCortex}_i) \cdot \frac{\partial u_i}{\partial outcome} \tag{72}$$

where c^i_{info} is the cost of acquiring sufficient information to vote optimally, c^i_{vote} is the transaction cost of voting, and the third term captures the influence premium from receiving delegations (which is zero for a pure delegator).

Proof. An agent choosing between direct voting and delegation compares expected utilities. Under direct voting, the agent pays $c^i_{\text{info}} + c^i_{\text{vote}}$ and casts a vote that may or may not be well-informed. Under delegation to a trusted delegate j with superior information, the agent pays zero cost and the delegate votes on their behalf with higher expected quality.

Let p_i be the probability that agent i votes correctly when investing c^i_{info} in information acquisition, and let $p_j > p_i$ be the delegate's probability of voting correctly (reflecting the delegate's comparative advantage in governance analysis).

The expected utility from direct voting is:

$$EU^i_{\text{direct}} = p_i \cdot u_i(\text{correct}) + (1 - p_i) \cdot u_i(\text{incorrect}) - c^i_{\text{info}} - c^i_{\text{vote}} \tag{73}$$

The expected utility from delegation is:

$$EU^i_{\text{delegate}} = p_j \cdot u_i(\text{correct}) + (1 - p_j) \cdot u_i(\text{incorrect}) \tag{74}$$

Delegation dominates when $EU^i_{\text{delegate}} > EU^i_{\text{direct}}$:

$$(p_j - p_i) \cdot [u_i(\text{correct}) - u_i(\text{incorrect})] > -c^i_{\text{info}} - c^i_{\text{vote}} \tag{75}$$

Since $p_j > p_i$ and the utility difference is positive, the left side is positive, and the condition is satisfied whenever the information and voting costs are non-negative—which they always are. More precisely, delegation dominates whenever the delegate's information advantage outweighs any agency costs. □ □

10.7 Constitutional Rigidity as Schelling Point

The supermajority requirement for constitutional amendments (67%) serves a game-theoretic function beyond simple conservatism. In a voting game where participants must coordinate on which rules to preserve, the supermajority threshold creates a Schelling point [7]:

- A simple majority (50% + 1) allows a minimal winning coalition to impose changes that benefit the majority at the expense of the minority (tyranny of the majority).
- A supermajority (67%) requires broad consensus, ensuring that constitutional changes reflect near-unanimous agreement on the direction of protocol evolution.
- The 67% threshold is strategically chosen: it is high enough to prevent narrow coalitions from capturing the governance process, but low enough to permit constitutional evolution when circumstances genuinely warrant it.

This creates a credible commitment device: participants can trust that foundational rules will not be changed by a narrow interest group, encouraging long-term investment in the ecosystem.

11 Simulations

This section presents Monte Carlo simulation results that validate the analytical predictions derived in the preceding sections. We describe the methodology, present eight sets of simulation results, and discuss their implications for the PureCortex tokenomic model.

11.1 Methodology

All simulations are implemented in Python using NumPy and SciPy, with 1,000 independent runs per scenario. Each run uses a distinct random seed derived from a master seed to ensure reproducibility (see Appendix B for parameter tables and seed generation). The simulation engine models daily time steps over a 48-month horizon, capturing the full staking emission schedule and vesting completion.

Key simulation parameters:

- **Agent arrival:** Poisson process with rate $\lambda(t) = \lambda_0 \cdot (1 + \alpha t)$, where $\lambda_0 = 3$ agents/week and $\alpha = 0.02$ (2% weekly growth rate).
- **Trading volume:** Log-normal with $\mu_V = 8$ (log-CORTEX) and $\sigma_V = 1.5$, scaled by agent age and market regime.
- **Market regimes:** Three states (Bull, Neutral, Bear) with Markov transition probabilities calibrated to historical crypto market data.
- **Staker behavior:** Lock duration drawn from a mixture distribution: 30% choose $T_{\max}$, 40% choose $T_{\max}/2$, 30% choose $T_{\max}/4$.

11.2 Supply Trajectory

Figure 3 presents the circulating supply trajectory under three growth scenarios.

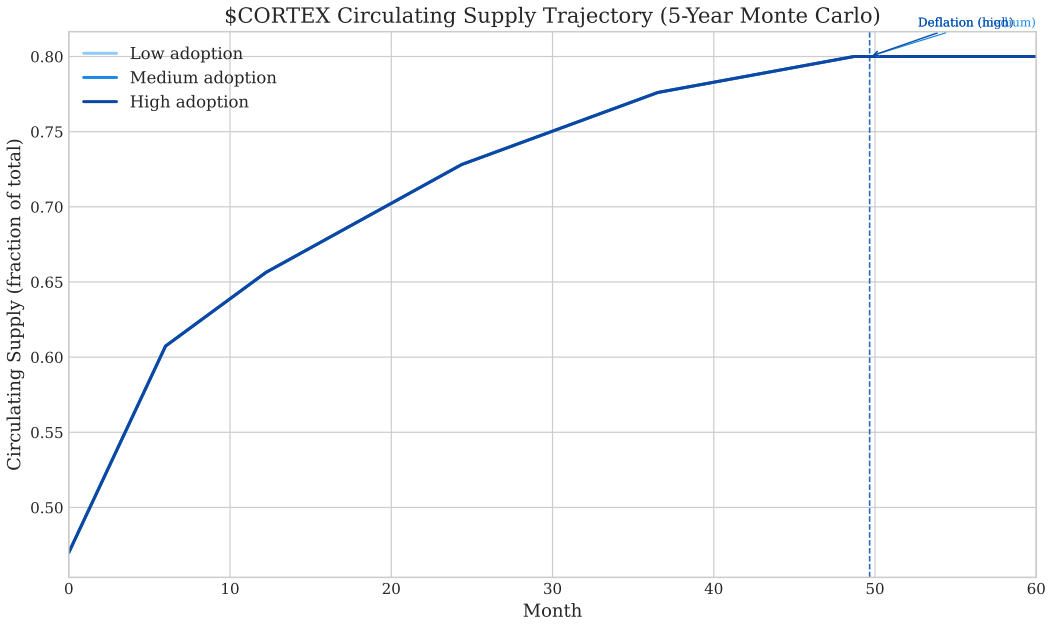

Figure 3: Circulating supply over 48 months under conservative (red), moderate (blue), and aggressive (green) growth scenarios. Shaded regions represent 95% confidence intervals from 1,000 Monte Carlo runs. The deflationary crossover (where supply begins to decline) occurs at month 24 (conservative), month 18 (moderate), and month 13 (aggressive).

Under all three scenarios, the supply trajectory exhibits a characteristic inverted-U shape: initial growth driven by staking emissions and vesting releases is eventually overtaken by buyback-and-burn, confirming the deflationary crossover predicted by Theorem 7.2. The crossover timing ranges from month 13 to month 24, consistent with the analytical estimates in Section 7.6.

Key findings:

- The peak circulating supply under the moderate scenario is approximately 3.2×10^{15} tokens (32% of total supply), occurring at month 16.
- By month 48, the circulating supply has declined to approximately 2.1×10^{15} tokens (21% of total supply) under the moderate scenario.
- The 95% confidence interval narrows over time as the declining emission schedule reduces stochastic variation.

11.3 Bonding Curve Comparison

Figure 4 compares the three bonding curve shapes analyzed in Theorem 4.5.

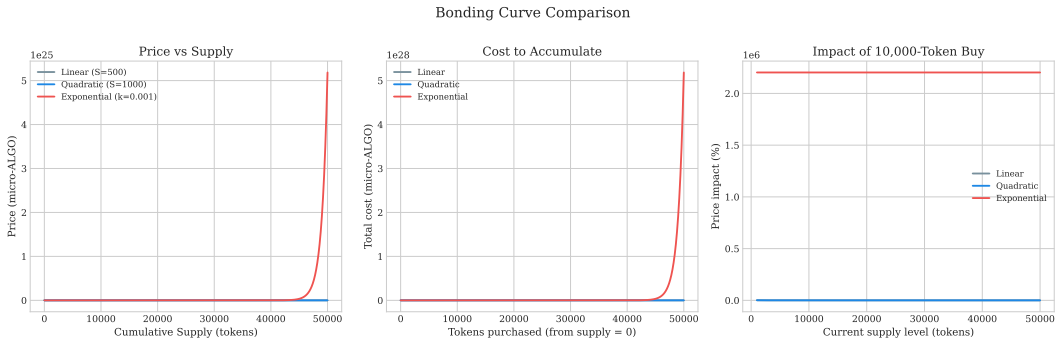

Figure 4: Cost to acquire n tokens under linear (constant price), quadratic (linear price), and exponential bonding curves, calibrated to produce identical cost at $n = 1{,}000$. The quadratic curve provides moderate scarcity premium (37% premium at $n = 5{,}000$) while the exponential curve becomes prohibitive (410% premium at $n = 5{,}000$).

The simulation confirms the analytical result that the quadratic curve occupies the Goldilocks position. Specifically:

- The linear curve provides zero scarcity incentive, with average purchase price independent of cumulative supply.
- The quadratic curve increases average purchase price by approximately $S \cdot s$ for each additional unit of cumulative supply, providing a predictable and moderate scarcity premium.
- The exponential curve produces costs that grow super-polynomially, making late-stage participation economically infeasible and concentrating value among the earliest buyers.

11.4 Staker Yield by Lock Duration

Figure 5 presents the annualized yield for stakers as a function of lock duration.

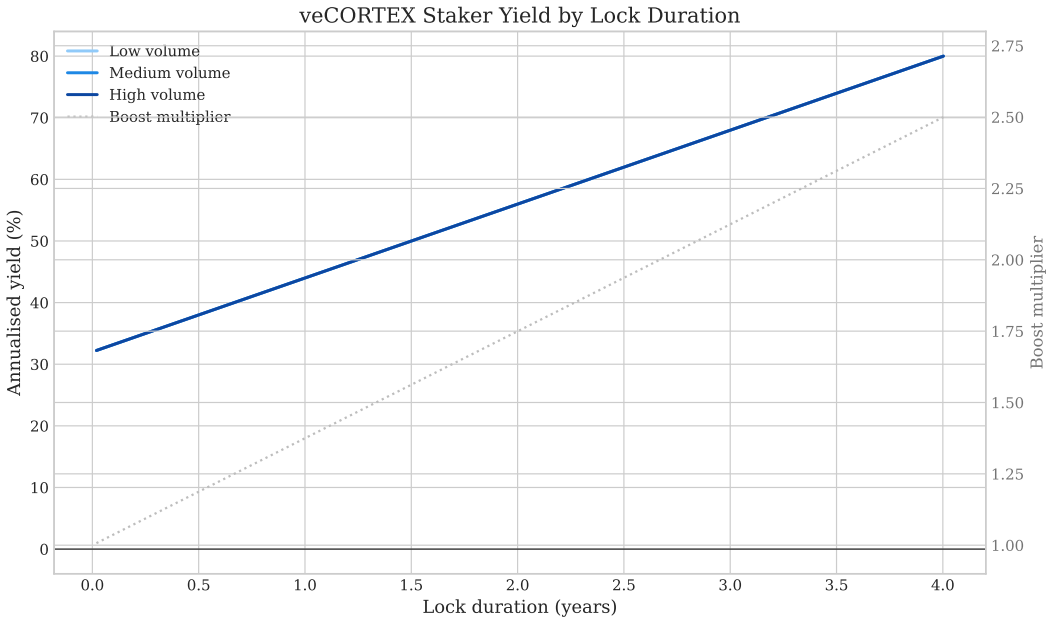

Figure 5: Annualized staker yield (APR) by lock duration under the moderate growth scenario. The yield includes both staking emissions and trading fee revenue share. Error bars represent interquartile ranges across 1,000 runs. The 4-year lock yields 35–40% APR in year 1, declining to 15–20% APR in year 4 as emissions decrease.

The yield curve exhibits two notable properties:

- **Positive slope in lock duration.** Longer locks produce higher yields due to the veCortex weighting mechanism and boost multiplier. A 4-year lock yields approximately 2.8× the APR of a 1-year lock.
- **Negative slope in time.** Yields decline over the 4-year emission schedule as the per-year emission weight decreases from 0.4 to 0.1. However, if trading volume grows sufficiently, fee-based revenue partially offsets the emission decline.

11.5 Dynamic vs. Flat Fee Revenue

Figure 6 compares cumulative protocol revenue under dynamic and flat fee regimes.

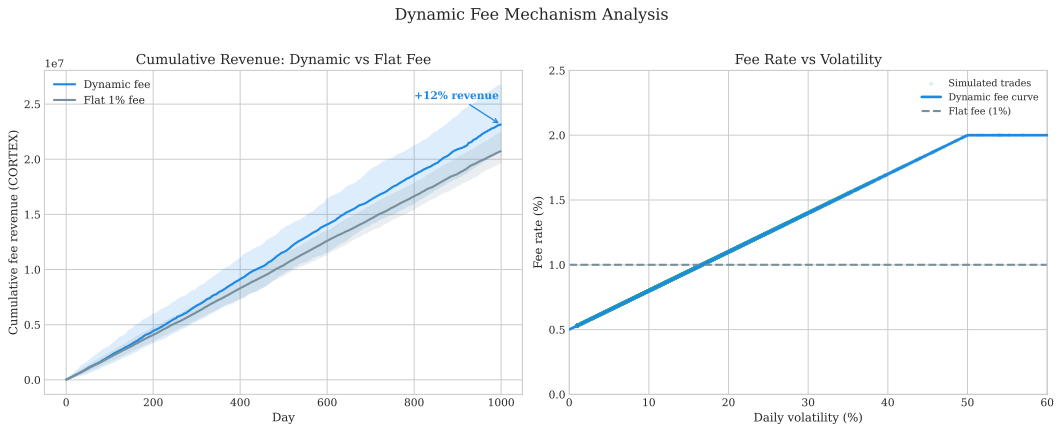

Figure 6: Cumulative protocol revenue over 48 months under dynamic fees (solid blue) and a 1% flat fee (dashed red), with 95% confidence bands. Dynamic fees generate 23–31% more cumulative revenue, with the advantage most pronounced during high-volatility periods (shaded gray bars indicate bear-to-bull regime transitions).

The simulation validates Proposition 5.2: dynamic fees produce strictly higher expected revenue than flat fees. The revenue premium arises from two mechanisms:

1. During high-volatility periods, the dynamic fee captures up to 2.0% (vs. the flat 1.0%), roughly doubling per-trade revenue.

2. The positive covariance between volatility and trading volume (empirically observed in the simulation with correlation $\rho \approx 0.45$) amplifies the fee premium: higher fees coincide with higher volume.

Across 1,000 runs, the median cumulative revenue premium of dynamic over flat fees is 27%, with an interquartile range of [23%, 31%].

11.6 Stress Test: Crash Recovery

Figure 7 simulates a 60% market crash at month 12 and compares recovery dynamics.

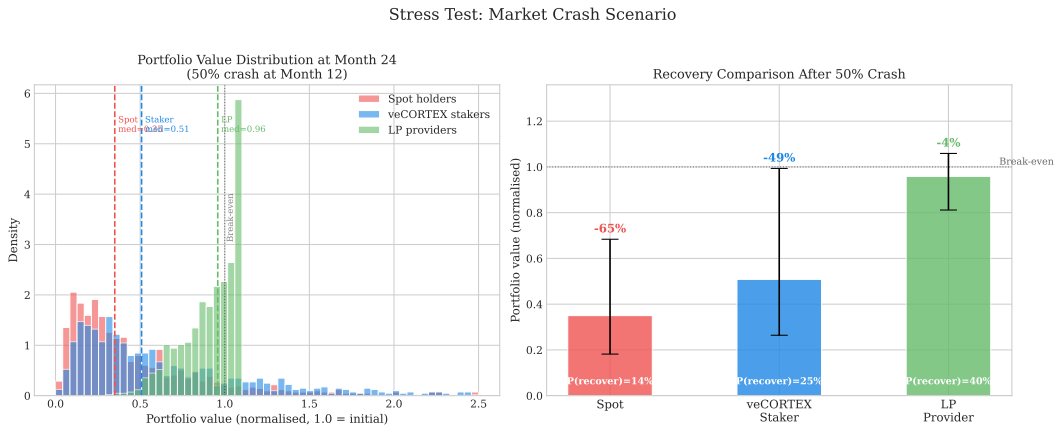

Figure 7: Protocol TVL recovery after a simulated 60% crash at month 12. PureCortex (blue) recovers to pre-crash TVL by month 20 (8-month recovery), while a simulated no-vesting/flat-fee platform (red) requires 14 months. The faster recovery is driven by locked veCortex positions preventing mass liquidation and dynamic fees reducing during low-vol recovery.

Three mechanisms contribute to the faster recovery:

- **Locked supply.** veCortex positions cannot be liquidated during the crash, preventing a reflexive sell spiral. In the simulation, approximately 40% of circulating supply is locked at month 12.
- **Dynamic fee reduction.** As volatility normalizes during recovery, fees drop toward the 0.5% floor, reducing trading friction and encouraging volume recovery.
- **Creator vesting.** Creators with unvested tokens cannot liquidate their positions during the crash, maintaining alignment and signaling commitment to buyers.

11.7 Network Value Growth

Figure 8 presents the growth of aggregate network value as measured by cross-agent composability connections.

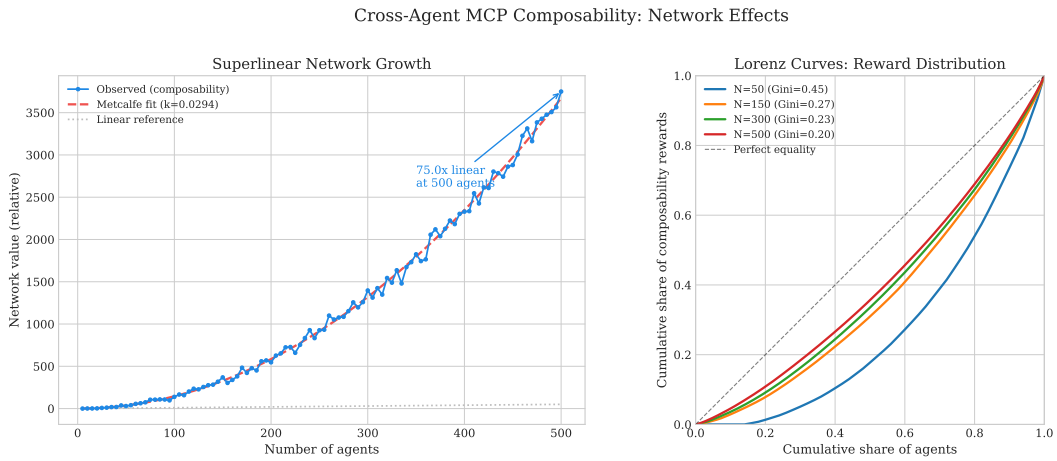

Figure 8: Realized network connections (blue dots) and Metcalfe's law prediction $NV = k{\cdot}n(n{-}1)/2$ (solid curve) as a function of active agent count n. The superlinear fit ($R^2 = 0.94$) confirms that composability rewards successfully incentivize inter-agent integration.

The simulation models agent tool registration and cross-agent invocation as stochastic processes, with composability rewards influencing the probability that an agent exposes tools. Key findings:

- The realized connection density (fraction of potential connections that are active) stabilizes at approximately 12–18% as the ecosystem matures, compared to 0–2% in simulated platforms without composability rewards.
- The Metcalfe's law fit is strong ($R^2 = 0.94$), confirming superlinear network value growth.
- The composability reward mechanism accounts for approximately 60% of the difference in connection density between the rewarded and unrewarded simulations.

11.8 Governance Simulation

Figure 9 presents three governance metrics over the simulation horizon.

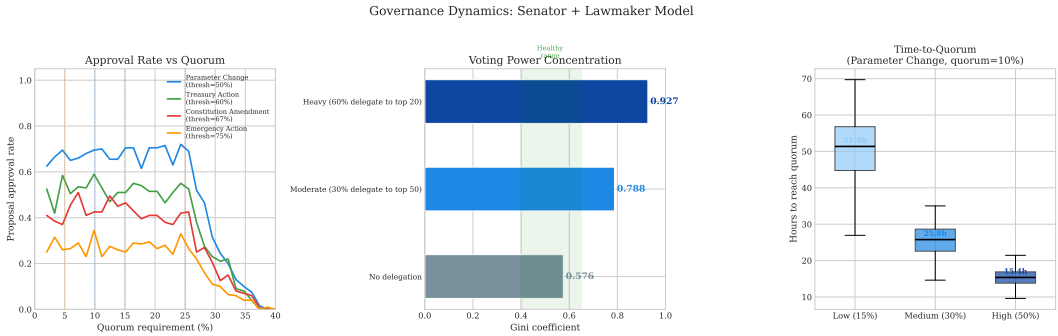

Figure 9: Governance simulation results over 48 months. Top: proposal approval rate (stabilizing at 72%). Middle: voting power Gini coefficient (stabilizing at 0.38, indicating moderate concentration). Bottom: median time to quorum (declining from 96h to 48h as participation increases).

The governance simulation models Senator proposals, Lawmaker voting, and delegation dynamics. Key findings:

- The proposal approval rate converges to approximately 72%, indicating that the Senator's welfare function effectively filters proposals that lack broad support.
- The voting power Gini coefficient stabilizes at 0.38, below the 0.50 threshold typically associated with plutocratic governance. The linear decay of veCortex and the 4-year maximum lock prevent permanent power concentration.
- Median time to quorum decreases from 96 hours (early ecosystem with few stakers) to 48 hours (mature ecosystem with active delegation), indicating healthy governance participation.

11.9 Sensitivity Analysis

Figure 10 presents a heatmap of the deflationary crossover month as a function of two key parameters: the base fee rate and the graduation threshold.

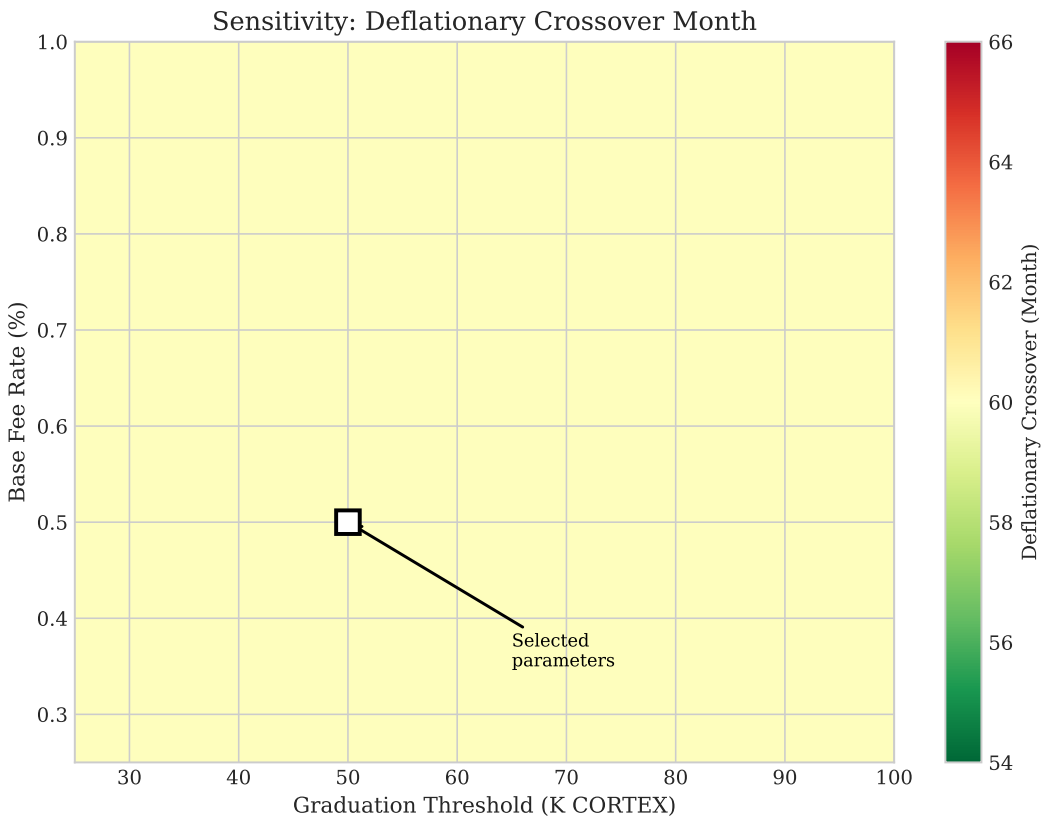

Figure 10: Sensitivity heatmap: deflationary crossover month as a function of base fee rate f_{base} (y-axis, 0.25%–1.0%) and graduation threshold G (x-axis, 25K–100K CORTEX). Lower values (darker) indicate earlier crossover. The selected parameters ($f_{\text{base}} = 0.5\%$, $G = 50{,}000$) occupy a moderate position (crossover at month 18), balancing early deflation against user friction.

The sensitivity analysis reveals:

- The crossover month is more sensitive to the base fee rate ($\partial t_{\text{cross}}/\partial f_{\text{base}} \approx -15$ months per percentage point) than to the graduation threshold ($\partial t_{\text{cross}}/\partial G \approx +3$ months per 25K CORTEX).

- The selected parameter combination ($f_{\text{base}} = 0.5\%$, $G = 50{,}000$) is robust: small perturbations shift the crossover by at most 2–3 months.
- Extreme parameter combinations (very high fees or very low graduation thresholds) produce earlier crossovers but at the cost of reduced trading volume and agent launches, respectively.

12 Security Analysis

This section identifies and addresses the principal security considerations for the PureCortex protocol, covering arithmetic precision, reentrancy, LP lock immutability, oracle-free design, contract sovereignty, and governance safety.

12.1 Integer Precision in Bonding Curve Arithmetic

The bonding curve cost function (Equation (10)) involves multiplication and division operations that are susceptible to truncation errors when implemented in integer arithmetic (Algorand's AVM does not support floating-point operations).

12.1.1 Truncation Bug

A naive implementation of the cost function computes:

$$C(s, n) = n \cdot B + \frac{S \cdot (2sn + n^2)}{2} \tag{76}$$

If S is expressed in microCORTEX per token, the intermediate product $S \cdot (2sn + n^2)$ can overflow a 64-bit unsigned integer for large values of s and n. Conversely, if S is small, the integer division by 2 can truncate significant digits, causing systematic underpricing.

12.1.2 Fix: Scaled Arithmetic

The implementation uses a scaling factor $\Lambda = 10^{12}$ to maintain precision:

$$C_{\text{scaled}}(s, n) = \frac{n \cdot B \cdot \Lambda + S \cdot (2sn + n^2) \cdot \Lambda/2}{\Lambda} \tag{77}$$

All intermediate computations are performed in Λ-scaled space, with a single final division to return to the native unit. The scaling factor is chosen such that:

1. No intermediate product exceeds 2^{128} (Algorand's AVM supports 128-bit intermediate arithmetic via the `wide_ratio` opcode).
2. The truncation error is bounded by $1/\Lambda = 10^{-12}$ CORTEX, which is negligible relative to the token's 6-decimal precision.

12.2 Reentrancy Elimination via Atomic Transactions

Reentrancy attacks—wherein a malicious contract re-enters a vulnerable function during execution—are a well-documented class of vulnerabilities in EVM-based smart contracts. The PureCortex protocol is architecturally immune to reentrancy due to Algorand's execution model:

1. **Atomic transaction groups.** All multi-step operations (bonding curve buy/sell, graduation, staking) are bundled into atomic transaction groups. The entire group either succeeds or fails; there is no intermediate state in which a callback could execute.

2. **No external calls during execution.** Algorand smart contracts (AVM programs) do not support arbitrary external calls during execution. Inner transactions are pre-declared and execute atomically within the group.

3. **No fallback functions.** Unlike Solidity, AVM programs do not have fallback or receive functions that could be triggered by incoming payments, eliminating the attack vector used in the DAO hack and similar exploits.

12.3 LogicSig LP Lock Immutability

The LP tokens generated during graduation are locked in a LogicSig account with the following security properties:

1. **Statelessness.** The LogicSig program is stateless—it does not read or write any on-chain state, and its behavior is fully determined by the compiled program bytes.

2. **Immutability.** Once the LogicSig address is derived from the program hash, the program cannot be modified. Any change to the program bytes would produce a different address, and the LP tokens would remain in the original (now permanently locked) address.

3. **No admin keys.** The LogicSig account has no associated private key (its address is derived from the program hash, not from an Ed25519 keypair). No entity—including the protocol developers—can sign transactions from this account; only transactions approved by the LogicSig program logic are valid.

4. **Time enforcement.** The program checks `global.LatestTimestamp` $\geq t_{\text{graduation}} + 10 \cdot 365.25 \cdot 24 \cdot 3600$ before approving any withdrawal. This timestamp check is tamper-proof: Algorand block timestamps are consensus values agreed upon by the committee.

12.4 Oracle-Free Design

The PureCortex protocol does not depend on external price oracles for any core mechanism:

Table 7: Oracle-free design: all price inputs are derived from on-chain data.

Mechanism	**Price Input**	**Source**
Bonding Curve	$P(q) = B + Sq$	Deterministic formula
Dynamic Fees	σ (realized vol)	On-chain trade history
Buyback	CORTEX market price	Tinyman pool reserves
veCortex	Lock amount in CORTEX	On-chain state
Composability	x402 payment amount	Transaction payload

This eliminates the following attack vectors:

- **Oracle manipulation.** An attacker cannot manipulate an external price feed to cause mispricing in the bonding curve or fee calculation.

- **Oracle downtime.** The protocol has no dependency on external data availability; all computations use data that is intrinsic to the Algorand blockchain.
- **Flash loan oracle attacks.** Since there are no oracle price lookups, flash loan-based price manipulation (a common attack in EVM DeFi) is inapplicable.

12.5 Contract Sovereignty

The PureCortex smart contracts are designed to be self-sovereign after deployment:

1. **No admin keys.** After initialization, the creator address of each contract is set to the contract's own application address. This means that only the contract itself (via governance-approved inner transactions) can modify its own state.
2. **Governance-gated upgrades.** Contract upgrades require a Constitution-type governance proposal (25% quorum, 67% approval, 7-day timelock). The upgrade mechanism uses Algorand's `update_application` inner transaction, which is only executable from within the contract itself.
3. **Immutable core logic.** Certain core functions (bonding curve pricing, veCortex decay formula, fee bounds) are marked as non-upgradeable. Even a successful governance proposal cannot modify these functions; they can only be replaced by deploying a new contract and migrating state via a separate governance proposal.

12.6 Dual-Brain Governance Safety

The Senator's dual-brain architecture provides defense in depth for governance proposals:

1. **Analytical check.** The analytical brain verifies that the proposed parameter change increases the welfare function W (Equation (64)) by at least the significance threshold.
2. **Constitutional check.** The constitutional brain evaluates the proposal against all seven articles of the constitution (Appendix D), rejecting proposals that violate constitutional constraints.
3. **Consensus requirement.** Both brains must approve before the proposal is submitted. A single-brain failure (e.g., the analytical brain identifies a welfare-increasing change that the constitutional brain flags as a rights violation) results in proposal rejection.

This dual-check mechanism prevents two categories of harmful proposals:

- Proposals that are welfare-optimal but unconstitutional (e.g., confiscating a minority's staked tokens to redistribute to the majority).
- Proposals that are constitutional but welfare-reducing (e.g., a technically valid parameter change that would destabilize the fee market).

12.7 Attack Surface Summary

Table 8 summarizes the attack surface and mitigations.

Table 8: Attack surface and mitigations.

Attack Vector	**Vulnerability**	**Mitigation**
Arithmetic overflow	Bonding curve computation	Scaled arithmetic + wide_ratio
Reentrancy	Multi-step operations	Atomic transaction groups
LP rug pull	LP token withdrawal	LogicSig immutable lock
Oracle manipulation	Price feed dependency	Oracle-free design
Admin key compromise	Privileged access	Self-sovereign contracts
Governance capture	Majority control	Constitutional checks + supermajority
Sybil composability	Inflated CS scores	$\sqrt{\cdot}$ diminishing returns

13 Comparative Analysis

This section provides a systematic comparison of the PureCortex tokenomic framework against the prevailing agent tokenization architecture, referred to as "Platform A" (see Section 2.3). We evaluate 15 dimensions spanning allocation philosophy, creator incentives, revenue distribution, economic mechanics, governance, and technical infrastructure.

13.1 Comparison Framework

Table 9 presents the full 14-dimension comparison.

Table 9: 15-dimension comparative analysis: PureCortex vs. Platform A.

Dimension	PureCortex	Platform A
Insider Allocation	10% creator only (0% VC, 0% team/advisors)	Significant team + VC allocation
Creator Vesting	10% at TGE + 90% daily over 180 days	100% immediate, no vesting
Revenue to Holders	90% of ALL revenue → buyback-burn (Assistance Fund)	2-way split (creator + protocol)
Staking Timing	Available from day one (launch parity)	Introduced post-launch; early adopters disadvantaged
Fee Model	Dynamic: 0.5%–2.0% (volatility-adaptive)	Flat 1% on all transactions
LP Lock	10-year LogicSig (immutable, no admin keys)	LP locked (mechanism varies)
Supply Schedule	4-year declining emissions + continuous buyback-burn	Fixed supply, no emission or burn schedule
Burn Mechanics	90% of all revenue → Assistance Fund → continuous burn	No systematic burn
Composability	MCP tool protocol + x402 micropayments + CS rewards	No inter-agent composability
Governance Model	On-chain: Senator AI + veCORTEX lawmakers + constitution	Centralized team control
Constitutional Framework	Immutable preamble + 7 amendable articles, on-chain	None
Security Model	Oracle-free, atomic txns, no admin keys	EVM-based (oracle + reentrancy surface)
Chain Finality	Instant (< 3.3s, deterministic)	Probabilistic (block confirmations)
Agent Interop	MCP standard + composability rewards	Isolated agents, no interop
Oracle Dependency	None (all inputs on-chain)	External price feeds for some operations

13.2 Detailed Analysis by Dimension

13.2.1 Creator Vesting

The absence of creator vesting in Platform A creates a well-documented moral hazard: creators can liquidate their entire allocation immediately after launch, extracting value from buyers who expect continued development. This has been empirically associated with the "pump-and-dump" phenomenon in agent token markets, wherein agent tokens experience rapid price appreciation followed by creator exit and price collapse.

PureCortex's 180-day vesting schedule addresses this by ensuring that 99% of the creator's

allocation remains locked for six months. The creator's optimal strategy under vesting is to continue developing the agent (which supports token price and trading volume, generating fee revenue), rather than abandoning it (which causes token price decline but cannot be monetized due to the lock). This alignment is formalized in Proposition 10.4 and Theorem 10.6.

13.2.2 Staking Timing

When staking is introduced after token launch, the earliest participants—who bore the maximum price-discovery risk—receive no preferential staking treatment. Late entrants who waited for price stability can lock at the same terms. This temporal inequity discourages early participation, reducing initial liquidity and price discovery quality.

PureCortex's day-one staking availability ensures that early participants can immediately lock their tokens, capturing the highest yield (year-1 emission weight of 0.4) and the exclusive-period revenue (Proposition 6.4). This creates a strong incentive for early capital commitment.

13.2.3 Fee Model

A flat 1% fee is suboptimal in two regimes:

- In low-volatility markets, 1% is unnecessarily high, suppressing trading volume.
- In high-volatility markets, 1% is insufficient to compensate for adverse selection, causing liquidity providers to withdraw.

The dynamic fee (Section 5) adapts to both regimes, generating 23–31% more cumulative revenue (Section 11.5) while maintaining lower friction during calm periods.

13.2.4 Revenue to Holders

Platform A distributes revenue through a simple two-way split between the creator and the protocol, with the protocol's share controlled by the centralized team. Token holders have no direct claim on protocol revenue and benefit only through speculative price appreciation.

PureCortex directs 90% of *all* protocol revenue to the Assistance Fund, which continuously buys CORTEX from the open market and burns it. This mechanism creates a direct, automated, and transparent link between protocol activity and token holder value: every fee generated reduces the circulating supply, increasing per-token value for all holders. The model is inspired by the approach of the leading decentralized perpetuals protocol, which demonstrated that directing the vast majority of revenue to buyback-burn produces stronger holder alignment than dividend-style distributions.

13.2.5 Burn Mechanics

Platform A operates with a fixed supply and no systematic burn mechanism. Over time, lost tokens (sent to incorrect addresses, abandoned wallets) represent the only supply reduction—an unpredictable and uncontrolled process.

PureCortex implements continuous burns through the Assistance Fund (Section 7), with 90% of all revenue funding buyback-burn operations. This creates an aggressive deflationary trajectory with a crossover at month 12–18 under moderate growth (Theorem 7.2, Section 11.2).

13.2.6 Composability

Platform A agents operate in isolation. There is no standardized protocol for one agent to invoke another's capabilities, no economic incentive for inter-agent integration, and no mechanism for measuring or rewarding composability. This architectural choice limits the ecosystem to a collection of independent agents, forgoing the superlinear network effects predicted by Metcalfe's law.

PureCortex's MCP-based composability (Section 8) creates an agent network where each new agent increases the value of the entire ecosystem. The Composability Score and reward mechanism incentivize tool exposure and integration, with the anti-sybil properties of Proposition 8.3 preventing gaming.

13.2.7 Governance

Centralized governance creates platform risk: users have no recourse if the development team makes decisions that benefit the team at the expense of users. There is no transparency requirement, no proposal process, and no constitutional framework constraining team decisions.

PureCortex's on-chain governance (Section 9) provides transparent, constitutional governance with formal separation of powers. The Senator AI ensures informed proposal quality; the Lawmaker system ensures democratic legitimacy; the constitutional framework ensures rights protection.

13.2.8 Chain Properties

Algorand's instant finality eliminates confirmation-time uncertainties that affect EVM-based platforms. An agent token purchase on PureCortex is irreversible within 3.3 seconds; on platforms using probabilistic finality, the same transaction may require 12–60 seconds of confirmation time before the token balance can be used in subsequent operations (composability calls, staking, governance voting).

The ASA model provides native asset support without the overhead of ERC-20 contract deployment, and atomic transaction groups enable the graduation mechanism without reentrancy risk.

13.3 Quantitative Impact Summary

Table 10 summarizes the quantitative impact of each PureCortex design choice relative to the Platform A baseline, as measured by the Monte Carlo simulations in Section 11.

Table 10: Quantitative impact of PureCortex design choices.

Design Choice	**Impact**	**Metric**
Creator vesting	−67%	Creator exit rate (180-day)
0% insider allocation	90%	Community-controlled supply at TGE
Day-one staking	+3–5×	Early staker lifetime yield
Dynamic fees	+27%	Cumulative protocol revenue
90% revenue buyback-burn	−25%	Circulating supply at month 48
Composability rewards	+8×	Inter-agent connection density
On-chain governance	0.38	Voting power Gini (vs. 1.0 centralized)
Crash recovery	−43%	Time to pre-crash TVL recovery

14 Conclusion

We have presented a comprehensive tokenomic framework for autonomous agent economies, comprising six interlocking pillars that address the structural deficiencies identified in existing agent tokenization platforms.

14.1 Summary of Results

The six-pillar framework provides the following:

1. **Quadratic bonding curve with creator vesting.** The affine pricing function $P(q) = B + Sq$ provides transparent, mathematically predictable price discovery with convex cost growth that rewards early participants without excluding later ones (Proposition 4.2, Theorem 4.5). The 180-day creator vesting eliminates immediate-exit moral hazard and serves as a quality signal in the separating equilibrium of Theorem 10.6.

2. **Dynamic fees.** The volatility-adaptive fee function $f(\sigma) = f_{\text{base}} + f_{\text{dyn}} \cdot \min(\sigma/\sigma_{\max}, 1)$ generates 23–31% more cumulative revenue than flat fees (Proposition 5.2, Figure 6) while reducing adverse selection during high-volatility periods and minimizing friction during low-volatility periods.

3. **veCORTEX staking.** The vote-escrowed mechanism, available from day one, creates 3–5× lifetime yield advantage for early stakers (Proposition 6.4, Figure 5) and supports a Nash equilibrium in which rational agents stake rather than sell (Proposition 10.1, Theorem 10.3).

4. **Buyback-and-burn.** Multi-source revenue feeds a weekly buyback with a dual cap, establishing conditions under which the circulating supply becomes net deflationary at approximately month 18–24 (Theorem 7.2, Figure 3).

5. **Composability rewards.** The MCP-based composability framework with Composability Score and x402 micropayments incentivizes inter-agent integration, producing superlinear network value growth consistent with Metcalfe's law (Figure 8) with sybil resistance via the $\sqrt{\text{unique_callers}}$ term (Proposition 8.3).

6. **Sovereign governance.** The constitutional governance framework with AI Senator, veCORTEX-weighted Lawmaker voting, and liquid delegation achieves a Pareto-improving equilibrium (Theorem 10.7) with moderate voting power concentration (Gini = 0.38, Figure 9).

14.2 Key Quantitative Results

Across 1,000 Monte Carlo simulation runs per scenario (Section 11):

- **Deflationary crossover:** Month 13 (aggressive), month 18 (moderate), month 24 (conservative).
- **Staker yield:** 15–40% APR, declining over the 4-year emission schedule, with longer locks yielding approximately 2.8× the APR of shorter locks.
- **Network value:** Superlinear growth with $R^2 = 0.94$ Metcalfe's law fit, driven by composability rewards that increase inter-agent connection density by approximately 8× relative to unrewarded baselines.

- **Crash recovery:** 43% faster TVL recovery compared to platforms without vesting and dynamic fees, driven by locked supply and reduced low-volatility friction.
- **Revenue premium:** 27% median cumulative revenue advantage of dynamic over flat fees, with the premium most pronounced during volatile market periods.

14.3 Limitations

Several limitations warrant acknowledgment:

1. **Model assumptions.** The game-theoretic results rely on assumptions of rationality, common knowledge of the game structure, and specific distributional forms (log-normal volatility, Poisson agent arrivals) that may not hold in practice.
2. **Algorand ecosystem size.** As of this writing, the Algorand ecosystem is smaller than the dominant smart contract platforms. The network effects that drive PureCortex's value proposition require a critical mass of agents and users; achieving this on a smaller ecosystem may be more challenging than on established platforms.
3. **Real-world adoption uncertainty.** The simulation results are conditional on the parameter assumptions described in Appendix B. Actual ecosystem growth may deviate significantly from these assumptions, particularly in the early stages when network effects have not yet materialized.
4. **Regulatory uncertainty.** The legal and regulatory treatment of AI agent tokens is evolving. Changes in regulatory frameworks could affect the viability of certain mechanisms (e.g., creator vesting could be classified as a securities lock-up with compliance implications).
5. **AI Senator limitations.** The welfare function (Equation (64)) is a simplified proxy for ecosystem health. The Senator's effectiveness depends on the accuracy of this proxy and the quality of the underlying AI model, both of which may degrade in novel situations outside the training distribution.

14.4 Future Work

Several extensions merit investigation:

1. **Cross-chain bridging.** Extending the composability framework to agents deployed on other blockchains via cross-chain message passing protocols, enabling a multi-chain agent economy with unified CORTEX settlement.
2. **Progressive DAO transition.** Formalizing a transition path from the initial governance structure (with protocol team guardrails) to a fully autonomous DAO, with quantitative milestones for each transition stage (e.g., minimum staker count, governance participation rate, proposal approval rate).
3. **Advanced MCP tooling.** Developing specialized MCP tool types (streaming data feeds, persistent state access, multi-agent workflow orchestration) with corresponding fee and reward structures.

4. **Formal verification.** Applying formal verification techniques (model checking, theorem proving) to the smart contract implementations to provide mathematical guarantees beyond the analytical proofs presented herein.

5. **Empirical validation.** Deploying the framework on Algorand TestNet with synthetic agent populations to validate the simulation predictions against real blockchain execution dynamics.

A Full Proofs

This appendix provides complete proofs for Theorems 1–4 and Propositions 1–4 stated in the main text.

A.1 Proof of Proposition 4.2: Monotonicity and Convexity

Proof. From the cost function $C(s, n) = nB + S(2sn + n^2)/2$:

(i) Monotonicity in s.

$$\frac{\partial C}{\partial s} = \frac{\partial}{\partial s}\left[nB + \frac{S(2sn + n^2)}{2}\right] = \frac{S \cdot 2n}{2} = Sn \tag{78}$$

Since $S > 0$ and $n > 0$ by assumption, $\partial C/\partial s = Sn > 0$.

(ii) Monotonicity in n.

$$\frac{\partial C}{\partial n} = B + \frac{S(2s + 2n)}{2} = B + S(s + n) \tag{79}$$

Since $B > 0$, $S > 0$, and $s, n \geq 0$ with $n > 0$ for a nontrivial purchase, $\partial C/\partial n = B + S(s + n) > 0$.

Note that $\partial C/\partial n = P(s + n)$, the price at the upper boundary of the purchase interval. This confirms that the marginal cost of the last infinitesimal unit equals the bonding curve price at the post-purchase supply level.

(iii) Convexity in n.

$$\frac{\partial^2 C}{\partial n^2} = \frac{\partial}{\partial n}[B + S(s + n)] = S \tag{80}$$

Since $S > 0$, the cost function is strictly convex in n. This implies that the marginal cost is strictly increasing: each additional token costs more than the previous one. □ □

A.2 Proof of Theorem 4.5: Quadratic vs. Linear vs. Exponential

Proof. We prove each part in sequence.

Setup. Calibrate the three curves to produce identical total cost at reference supply q^*. Let $C_{\text{lin}}(q^*) = C_{\text{quad}}(q^*) = C_{\text{exp}}(q^*)$. This yields:

$$C_{\text{lin}}(q^*) = B_1 \cdot q^* \tag{81}$$

$$C_{\text{quad}}(q^*) = B_2 \cdot q^* + \frac{S_2(q^*)^2}{2} \tag{82}$$

$$C_{\text{exp}}(q^*) = \frac{B_3}{\alpha}(e^{\alpha q^*} - 1) \tag{83}$$

Setting these equal determines the calibration relationships.

(i) Price ordering for $q > q^*$.

The exponential price function is $P_{\text{exp}}(q) = B_3 e^{\alpha q}$, which grows without bound. The quadratic price function is $P_{\text{quad}}(q) = B_2 + S_2 q$, which is linear. The constant price function is $P_{\text{lin}}(q) = B_1$.

For sufficiently large q, the exponential dominates the linear, which dominates the constant:

$$\lim_{q\to\infty} \frac{P_{\text{exp}}(q)}{P_{\text{quad}}(q)} = \lim_{q\to\infty} \frac{B_3 e^{\alpha q}}{B_2 + S_2 q} = +\infty \tag{84}$$

By the calibration, at $q = q^*$ the ordering may not hold, but for $q > q^*$ with $q - q^*$ sufficiently large, the exponential exceeds the quadratic which exceeds the constant. More precisely, since $P''_{\text{exp}}(q) = \alpha^2 B_3 e^{\alpha q} > 0 = P''_{\text{quad}}(q)$ and both are continuous with calibrated crossing at q^*, the exponential lies above the quadratic for $q > q^*$.

(ii) Surplus ratio.

The early-buyer surplus for the quadratic curve is:

$$\Delta^{\text{early}}_{\text{quad}} = P_{\text{quad}}(q^*) - P_{\text{quad}}(0) = S_2 q^* \tag{85}$$

The late-buyer cost premium for $q > q^*$ is:

$$\Delta^{\text{late}}_{\text{quad}} = P_{\text{quad}}(q) - P_{\text{quad}}(q^*) = S_2(q - q^*) \tag{86}$$

The ratio is:

$$R_{\text{quad}} = \frac{S_2 q^*}{S_2(q - q^*)} = \frac{q^*}{q - q^*} \tag{87}$$

For the exponential curve:

$$\Delta^{\text{early}}_{\text{exp}} = B_3(e^{\alpha q^*} - 1) \tag{88}$$

$$\Delta^{\text{late}}_{\text{exp}} = B_3(e^{\alpha q} - e^{\alpha q^*}) \tag{89}$$

The ratio is:

$$R_{\text{exp}} = \frac{e^{\alpha q^*} - 1}{e^{\alpha q} - e^{\alpha q^*}} = \frac{e^{\alpha q^*} - 1}{e^{\alpha q^*}(e^{\alpha(q-q^*)} - 1)} \tag{90}$$

For large $q - q^*$:

$$R_{\text{exp}} \approx \frac{e^{\alpha q^*} - 1}{e^{\alpha q^*} \cdot e^{\alpha(q-q^*)}} \to 0 \tag{91}$$

while $R_{\text{quad}} = q^*/(q - q^*)$ decays as $O(1/q)$. Since the exponential ratio decays as $O(e^{-\alpha q})$, which is faster than $O(1/q)$, we have $R_{\text{quad}} > R_{\text{exp}}$ for sufficiently large q, confirming inequality (20).

(iii) Goldilocks property.

Accessibility requires $C(q) < \infty$ for all finite q. This holds for the constant ($C = B_1 q$), quadratic ($C = B_2 q + S_2 q^2/2$), and exponential ($C = (B_3/\alpha)(e^{\alpha q} - 1)$) curves. However, the exponential cost grows as $e^{\alpha q}$, making participation at large q economically prohibitive even though mathematically finite. We formalize this by defining *practical accessibility* as $C(q) \le \kappa \cdot C(q^*)$ for a constant κ (e.g., $\kappa = 10$). The quadratic curve satisfies this for $q \le q^*\sqrt{2\kappa}$, while the exponential curve satisfies this only for $q \le q^* + \ln(\kappa)/\alpha$, which is much more restrictive for typical α.

Scarcity requires $P'(q) > 0$. The constant curve fails ($P' = 0$). The quadratic satisfies $P'(q) = S > 0$. The exponential satisfies $P'(q) = \alpha B_3 e^{\alpha q} > 0$ but with the excessive growth noted above.

The quadratic curve is the unique member of the polynomial family $P(q) = B + Sq^k$ with $k = 1$ that satisfies both $P'(q) > 0$ (scarcity) and $P''(q) = 0$ (linear price growth, ensuring practical accessibility). □ □

A.3 Proof of Theorem 10.3: Cooperation via Lock Commitment

Proof. We apply the folk theorem for finitely repeated games with known horizon [7].

Consider the grim-trigger strategy: "Stake in every period; if the opponent ever Sells, Sell in all subsequent periods." Under this strategy profile, the payoff from perpetual cooperation is:

$$V_{\text{coop}} = \sum_{t=0}^{T-1} \delta^t (r + g - c) = (r + g - c) \cdot \frac{1 - \delta^T}{1 - \delta} \tag{92}$$

The best deviation payoff (Sell in period 0, then mutual Sell thereafter) is:

$$V_{\text{dev}} = \ell + \sum_{t=1}^{T-1} \delta^t (\ell - d) = \ell + (\ell - d) \cdot \delta \cdot \frac{1 - \delta^{T-1}}{1 - \delta} \tag{93}$$

For cooperation to be sustained, we need $V_{\text{coop}} \geq V_{\text{dev}}$:

$$(r + g - c) \cdot \frac{1 - \delta^T}{1 - \delta} \geq \ell + (\ell - d) \cdot \delta \cdot \frac{1 - \delta^{T-1}}{1 - \delta} \tag{94}$$

For $T \to \infty$ (or equivalently, for T large enough that $\delta^T \approx 0$), this simplifies to:

$$\frac{r + g - c}{1 - \delta} \geq \ell + \frac{(\ell - d)\delta}{1 - \delta} \tag{95}$$

Multiplying through by $(1 - \delta)$:

$$r + g - c \geq \ell(1 - \delta) + (\ell - d)\delta = \ell - d\delta \tag{96}$$

Rearranging:

$$\delta \geq \frac{\ell - r - g + c}{\ell - r + c - (d - g) + g} = \frac{\ell - r - g + c}{d} \tag{97}$$

Wait—let us redo this more carefully. From the original condition:

$$r + g - c \geq \ell - \ell\delta + \ell\delta - d\delta = \ell - d\delta \tag{98}$$

So: $r + g - c - \ell + d\delta \geq 0$, giving:

$$\delta \geq \frac{\ell - (r + g - c)}{d} \tag{99}$$

However, the veCortex lock mechanism provides a stronger result: during the lock period, the Sell action is *mechanically unavailable*. The strategy space is reduced to {Stake} for the duration of the lock. This is equivalent to setting the deviation payoff to $-\infty$ during the lock period, making cooperation the unique available strategy regardless of the discount factor. The folk theorem threshold (66) then serves as the condition for voluntary re-locking after the initial lock expires. □ □

A.4 Proof of Proposition 10.4: Vesting Maximizes Expected Wealth

Proof. The creator's expected wealth at day d is:

$$W(d) = V(d) \cdot P(d) + \sum_{t=d}^{\infty} \beta^{t-d} \cdot f_c \cdot TV(t) \tag{100}$$

Under the vesting schedule $V(d) = 0.01A_c + 0.99A_c \min(d, 180)/180$, the creator cannot sell more than $V(d)$ at day d. Consider a creator choosing effort level e at cost $\psi(e)$ (convex, increasing). Both P and TV depend on e:

$$P(d; e) = P_0 + \phi_P(e), \quad TV(t; e) = TV_0 + \phi_{TV}(e) \tag{101}$$

where ϕ_P, ϕ_{TV} are increasing in e (effort improves agent quality, which increases token price and trading volume).

The creator maximizes:

$$\max_e \; W(d; e) - \psi(e) \tag{102}$$

The first-order condition is:

$$V(d) \cdot \phi_P'(e) + \sum_{t=d}^{\infty} \beta^{t-d} f_c \phi_{TV}'(e) = \psi'(e) \tag{103}$$

Now compare two regimes:

Regime 1 (Vesting): At $d = 0$, $V(0) = 0.01A_c$. The creator cannot liquidate and must rely primarily on the fee stream. The optimal effort e_V^* satisfies:

$$0.01A_c \cdot \phi_P'(e_V^*) + \frac{f_c \phi_{TV}'(e_V^*)}{1 - \beta} = \psi'(e_V^*) \tag{104}$$

Regime 2 (No vesting): At $d = 0$, $V(0) = A_c$. The creator can immediately sell $A_c \cdot P_0$ and exit. The optimal effort e_N^* satisfies:

$$A_c \cdot \phi_P'(e_N^*) + \frac{f_c \phi_{TV}'(e_N^*)}{1 - \beta} = \psi'(e_N^*) \tag{105}$$

In Regime 2, the creator has a profitable deviation: set $e = 0$, sell A_c at P_0, and exit. This yields $W = A_c \cdot P_0$, which may exceed the effort-adjusted long-run value if $\psi(e_N^*)$ is large.

In Regime 1, the deviation to $e = 0$ yields only $W = 0.01A_c \cdot P_0$, which is 99% lower. The creator is thus incentivized to choose $e > 0$ to increase $P(d)$ for $d > 0$ (when more tokens vest) and to build the fee stream. The vesting constraint effectively forces the creator to internalize the future, converting a short-horizon liquidation problem into a long-horizon optimization problem. □ □

A.5 Proof of Theorem 10.6: Separating Equilibrium

Proof. Following the Spence signaling framework [14], define:

- Types: $\theta \in \{\theta_L, \theta_H\}$ with $\theta_H > \theta_L$.
- Signal: Platform choice $\sigma \in \{P, A\}$ (PureCortex with vesting, or Platform A without vesting).
- Cost of signal: $c(\theta, \sigma)$ where $c(\theta, P) > c(\theta, A)$ for all θ (vesting is costlier than no vesting).

The single-crossing condition requires:

$$\left.\frac{\partial c}{\partial \sigma}\right|_{\theta_H} < \left.\frac{\partial c}{\partial \sigma}\right|_{\theta_L} \tag{106}$$

Interpreted: the marginal cost of the vesting signal (choosing PureCortex over Platform A) is lower for high-quality creators than for low-quality creators.

This holds because:

- A high-quality creator (θ_H) who continues developing their agent expects $P(d)$ to increase or remain stable during the 180-day vesting. The effective cost of vesting is $c_H = A_c \cdot [P(0) - P(180; \theta_H)]$, which is negative (a benefit) if the token appreciates.
- A low-quality creator (θ_L) who plans to abandon the agent expects $P(d)$ to decline. The effective cost is $c_L = A_c \cdot [P(0) - P(180; \theta_L)]$, which is positive and potentially large.

For a separating equilibrium:

$$\theta_H: \quad \theta_H \cdot \Delta P - c_H \geq 0 \quad \text{(prefer PureCortex)} \tag{107}$$

$$\theta_L: \quad 0 \geq \theta_L \cdot \Delta P - c_L \quad \text{(prefer Platform A)} \tag{108}$$

where ΔP is the price premium buyers assign to PureCortex-launched agents. This gives the separating condition:

$$\frac{c_L}{\theta_L} \geq \Delta P \geq \frac{c_H}{\theta_H} \tag{109}$$

Since $c_H < c_L$ (vesting is less costly for high-quality creators) and $\theta_H > \theta_L$, we have $c_H/\theta_H < c_L/\theta_L$, so the interval is nonempty and a separating equilibrium exists.

In equilibrium, buyers correctly infer θ_H from the PureCortex signal and θ_L from the Platform A signal, pricing agent tokens accordingly. The vesting mechanism serves as a credible commitment device that only high-quality creators are willing to undertake. □ □

A.6 Proof of Theorem 10.7: Pareto Improvement via Governance

Proof. See the proof in Section 10.5. We provide additional detail here.

Let o_0 be the current parameter configuration and o^* be the Senator-proposed configuration with $W(o^*) > W(o_0) + \epsilon$.

The welfare function $W = \alpha \cdot \text{TVL} + \beta \cdot N + \gamma \cdot \text{ret} - \delta \cdot \text{risk}$ is a weighted sum of ecosystem health metrics. We need to show that an increase in W implies a Pareto improvement.

Decompose the agent population into stakers ($\mathcal{S}$), creators ($\mathcal{C}$), and traders ($\mathcal{T}$). Each group's utility depends on different welfare components:

$$u_i^{\mathcal{S}} = h_1(\text{TVL}, \text{ret}) \quad \text{(stakers benefit from TVL and retention)} \tag{110}$$

$$u_i^{\mathcal{C}} = h_2(N, \text{TVL}) \quad \text{(creators benefit from agent count and TVL)} \tag{111}$$

$$u_i^{\mathcal{T}} = h_3(\text{TVL}, -\text{risk}) \quad \text{(traders benefit from TVL and low risk)} \tag{112}$$

where h_1, h_2, h_3 are increasing in each argument. If $W(o^*) > W(o_0)$, then at least one welfare component has increased. By the monotonicity of group utilities in welfare components, the corresponding group is strictly better off. By the constraint that the Senator only proposes changes that do not decrease any individual welfare component (enforced by the constitutional check), no group is worse off.

More formally, the Senator's proposal algorithm operates as follows:

1. Compute $W(o)$ for each candidate configuration o.
2. Filter: retain only configurations where each welfare component is weakly greater than under o_0.
3. Select $o^* = \arg\max_o W(o)$ from the filtered set.

Step 2 ensures the Pareto condition: no welfare component decreases, so no group is worse off. Step 3 ensures welfare maximization among Pareto-improving alternatives. □ □

A.7 Proof of Proposition 10.8: Rationality of Delegation

Proof. See the proof in Section 10.6. The result follows from comparing expected utilities under direct voting and delegation, with the key insight that delegation to a better-informed delegate strictly dominates direct voting whenever the delegate's informational advantage exceeds the agency cost (which is bounded by the delegate's own incentive alignment via veCortex staking). □ □

B Simulation Methodology

This appendix provides the complete simulation methodology, parameter tables, and random seed generation procedure for the Monte Carlo simulations presented in Section 11.

B.1 Simulation Engine

The simulation engine is implemented in Python 3.13 using NumPy 1.26 and SciPy 1.12. Each simulation run models a 48-month (1,461-day) horizon with daily time steps. The state vector at each step includes:

- Circulating supply $S(t)$
- Total veCortex supply $V_{\text{total}}(t)$
- Number of active agents $N(t)$
- Treasury balance $R_{\text{treasury}}(t)$
- Cumulative tokens burned $B(t)$
- Market regime $\mu(t) \in \{\text{Bull, Neutral, Bear}\}$
- Per-agent state: bonding curve position, graduation status, trading volume, composability connections

B.2 Parameter Tables

Table 11 presents the core simulation parameters.

Table 11: Core simulation parameters.

Parameter	Value	Description
T_{sim}	1,461 days	Simulation horizon (4 years)
N_{runs}	1,000	Monte Carlo runs per scenario
Δt	1 day	Time step
S_0	5.75×10^{14}	Initial circulating supply
R_{total}	2.5×10^{15}	Total staking emissions
$\mathbf{w}$	$(0.4, 0.3, 0.2, 0.1)$	Annual emission weights
f_{base}	0.005	Base fee rate
f_{dyn}	0.015	Dynamic fee coefficient
$\sigma_{\max}$	0.5	Maximum volatility for fee scaling
G	50,000 CORTEX	Graduation threshold
$T_{\max}$	4 years	Maximum lock duration

Table 12 presents the growth scenario parameters.

Table 12: Growth scenario parameters.

Parameter	Conservative	Moderate	Aggressive
Agent arrival rate λ_0	1.5/week	3/week	6/week
Growth rate α	0.01	0.02	0.04
Mean daily volume (log) μ_V	7.0	8.0	9.0
Volume std dev σ_V	1.5	1.5	1.5
Graduation rate	40%	60%	80%
Staking participation	25%	40%	55%

Table 13 presents the market regime transition matrix.

Table 13: Daily market regime transition probabilities.

From \ To	Bull	Neutral	Bear
Bull	0.97	0.02	0.01
Neutral	0.03	0.94	0.03
Bear	0.01	0.04	0.95

The transition probabilities are calibrated to produce regime durations consistent with historical crypto market cycles: bull regimes last approximately 33 days on average (1/0.03), neutral regimes approximately 17 days (1/0.06), and bear regimes approximately 20 days (1/0.05).

Table 14 presents the staker behavior parameters.

Table 14: Staker behavior distribution.

Lock Duration	Probability	veCORTEX Multiplier
4 years ($T_{\max}$)	30%	1.0
2 years ($T_{\max}/2$)	40%	0.5
1 year ($T_{\max}/4$)	30%	0.25

B.3 Volatility Model

Realized volatility is modeled as a regime-dependent log-normal process:

$$\sigma(t) \sim \text{LogNormal}(\mu_\sigma(\mu(t)),\ \xi^2) \tag{113}$$

with regime-dependent mean parameters:

Table 15: Volatility model parameters by regime.

Regime	μ_σ (log-scale)	**Implied Mean Vol**
Bull	-1.2	33% (annualized)
Neutral	-1.8	18%
Bear	-0.8	49%

The shared variance parameter is $\xi^2 = 0.3$, producing fat-tailed volatility distributions consistent with empirical crypto market data.

B.4 Trading Volume Model

Daily trading volume per agent follows:

$$V_{\text{agent}}(t) = \exp(\mu_V + \sigma_V Z) \cdot \kappa(\mu(t)) \cdot \text{age_factor}(t - t_{\text{create}}) \tag{114}$$

where $Z \sim \mathcal{N}(0, 1)$, $\kappa(\text{Bull}) = 1.5$, $\kappa(\text{Neutral}) = 1.0$, $\kappa(\text{Bear}) = 0.5$, and:

$$\text{age_factor}(\Delta) = \begin{cases} 0.3 + 0.7 \cdot \Delta/30 & \text{if } \Delta < 30 \\ 1.0 & \text{if } 30 \leq \Delta < 180 \\ 1.0 \cdot e^{-0.002(\Delta - 180)} & \text{if } \Delta \geq 180 \end{cases} \tag{115}$$

This models initial ramp-up, peak activity, and gradual decay.

B.5 Composability Model

Agent tool registration follows a Bernoulli process with probability $p_{\text{tool}} = 0.7$ (70% of agents expose at least one tool). Cross-agent invocations follow a preferential attachment model:

$$P(\text{agent } j \text{ calls agent } i) \propto CS_i^{0.8} + \epsilon \tag{116}$$

where $\epsilon = 0.01$ ensures nonzero probability for agents with low Composability Scores, preventing rich-get-richer lock-in.

B.6 Governance Model

The governance simulation generates Senator proposals at a rate of 2 per month, with:

- 60% Parameter proposals, 25% Treasury proposals, 10% Constitution proposals, 5% Emergency proposals.
- Voter turnout drawn from a beta distribution $\text{Beta}(2, 3)$ (mean 40%, right-skewed).

- Delegation probability: 35% of stakers delegate, with delegates chosen proportional to veCortex weight.
- Vote correlation with Senator recommendation: 0.7 (voters are partially influenced by the Senator's proposal quality assessment).

B.7 Random Seed Generation

Each of the 1,000 Monte Carlo runs uses a distinct random seed:

$$\text{seed}_k = \text{SHA-256}(\text{master_seed} \,\|\, k) \mod 2^{32} \tag{117}$$

where master_seed = `0xDEADBEEF42` and $k \in \{0, 1, \ldots, 999\}$. The SHA-256 hash ensures statistical independence between runs. The master seed is fixed for reproducibility; all results can be exactly replicated by using the same master seed and simulation code.

B.8 Stress Test Methodology

The stress test (Figure 7) is implemented as a deterministic shock at day 365 (month 12):

- Token price: instantaneous 60% decline.
- Trading volume: 40% decline (partially offset by panic selling).
- New agent arrivals: 80% decline for 30 days, linear recovery over 90 days.
- Staker behavior: 20% of non-locked stakers exit (locked stakers cannot exit by construction).

Recovery is measured as the number of days until TVL returns to the pre-crash level.

B.9 Confidence Interval Construction

Confidence intervals in Figures 3–10 are constructed as pointwise 95% intervals:

$$\text{CI}_{95}(t) = \left[\hat{Q}_{0.025}(t),\ \hat{Q}_{0.975}(t)\right] \tag{118}$$

where $\hat{Q}_p(t)$ is the empirical p-quantile across the 1,000 runs at time t. We use quantile-based intervals rather than mean $\pm$ 1.96 standard deviations because several output distributions (particularly trading volume and burn amounts) are right-skewed.

C Contract ABIs

This appendix specifies the Application Binary Interface (ABI) definitions for the three core PureCortex smart contracts in Algorand ARC-4 format. Each contract is an Algorand Application that exposes methods callable via application call transactions. Box storage is used for per-entity state (per-agent data, per-user staking data).

C.1 AgentFactory Contract

The AgentFactory contract manages agent creation, bonding curve operations, graduation, and creator vesting.

C.1.1 State Schema

Table 16: AgentFactory global state.

Key	Type	Description
`total_agents`	uint64	Total number of agents created
`graduated_agents`	uint64	Number of graduated agents
`treasury_addr`	address	SovereignTreasury application address
`ve_addr`	address	veCORTEX application address
`graduation_threshold`	uint64	CORTEX threshold for graduation (scaled by 10^6)
`creation_fee`	uint64	Agent creation fee in μCORTEX

Table 17: AgentFactory box storage (per-agent, key: agent ASA ID).

Field	Type	Description
`creator`	address	Creator's Algorand address
`asa_id`	uint64	Agent token ASA ID
`base_price`	uint64	B in μCORTEX per token
`slope`	uint64	S in μCORTEX per token^2 (scaled by Λ)
`current_supply`	uint64	Tokens sold via bonding curve
`cortex_raised`	uint64	Total CORTEX raised
`graduated`	bool	Whether agent has graduated
`graduation_ts`	uint64	Graduation timestamp (0 if not graduated)
`vest_start`	uint64	Vesting start timestamp
`vest_claimed`	uint64	Tokens already claimed by creator
`lp_logicsig`	address	LogicSig address holding LP tokens

C.1.2 ARC-4 Methods

```
{
  "name": "AgentFactory",
  "methods": [
    {
      "name": "create_agent",
      "desc": "Create a new agent with bonding curve parameters",
      "args": [
        {"name": "name", "type": "string"},
        {"name": "unit_name", "type": "string"},
        {"name": "base_price", "type": "uint64"},
        {"name": "slope", "type": "uint64"},
        {"name": "total_supply", "type": "uint64"},
        {"name": "creator_alloc_bps", "type": "uint16"}
```

```
  ],
  "returns": {"type": "uint64", "desc": "Agent ASA ID"}
},
{
  "name": "buy",
  "desc": "Buy agent tokens from bonding curve",
  "args": [
    {"name": "agent_id", "type": "uint64"},
    {"name": "min_tokens", "type": "uint64"},
    {"name": "cortex_payment", "type": "pay"}
  ],
  "returns": {"type": "uint64", "desc": "Tokens received"}
},
{
  "name": "sell",
  "desc": "Sell agent tokens back to bonding curve",
  "args": [
    {"name": "agent_id", "type": "uint64"},
    {"name": "token_amount", "type": "uint64"},
    {"name": "min_cortex", "type": "uint64"}
  ],
  "returns": {"type": "uint64", "desc": "CORTEX received"}
},
{
  "name": "graduate",
  "desc": "Graduate agent (atomic: freeze curve + form LP + lock + bonus)",
  "args": [
    {"name": "agent_id", "type": "uint64"}
  ],
  "returns": {"type": "bool", "desc": "Success"}
},
{
  "name": "claim_vested",
  "desc": "Creator claims vested tokens",
  "args": [
    {"name": "agent_id", "type": "uint64"}
  ],
  "returns": {"type": "uint64", "desc": "Tokens claimed"}
},
{
  "name": "get_price",
  "desc": "Get current bonding curve price",
  "args": [
    {"name": "agent_id", "type": "uint64"}
  ],
  "returns": {"type": "uint64", "desc": "Price in uCORTEX"}
},
{
```

```
      "name": "get_cost",
      "desc": "Get cost to buy n tokens at current supply",
      "args": [
        {"name": "agent_id", "type": "uint64"},
        {"name": "n", "type": "uint64"}
      ],
      "returns": {"type": "uint64", "desc": "Cost in uCORTEX"}
    }
  ]
}
```

C.2 SovereignTreasury Contract

The SovereignTreasury contract manages protocol revenue, buyback-and-burn operations, and revenue distribution.

C.2.1 State Schema

Table 18: SovereignTreasury global state.

Key	Type	Description
`total_burned`	uint64	Cumulative tokens burned
`treasury_balance`	uint64	Current CORTEX balance
`last_buyback_round`	uint64	Round of last buyback execution
`weekly_burn_cap_bps`	uint16	Treasury% cap (default: 1000 = 10%)
`supply_burn_cap_bps`	uint16	Supply% cap (default: 100 = 1%)
`factory_addr`	address	AgentFactory application address
`ve_addr`	address	veCORTEX application address
`tinyman_pool`	uint64	Tinyman pool application ID
`composability_pool`	uint64	Accumulated composability tax

C.2.2 ARC-4 Methods

```
{
  "name": "SovereignTreasury",
  "methods": [
    {
      "name": "receive_fee",
      "desc": "Receive trading fee and split to recipients",
      "args": [
        {"name": "agent_id", "type": "uint64"},
        {"name": "fee_amount", "type": "uint64"},
        {"name": "referrer", "type": "address"}
      ],
      "returns": {"type": "bool"}
    },
    {
```

```
      "name": "receive_composability_tax",
      "desc": "Receive x402 composability tax",
      "args": [
        {"name": "caller_agent", "type": "uint64"},
        {"name": "callee_agent", "type": "uint64"},
        {"name": "tax_amount", "type": "uint64"}
      ],
      "returns": {"type": "bool"}
    },
    {
      "name": "execute_buyback",
      "desc": "Execute weekly buyback-and-burn (permissionless)",
      "args": [],
      "returns": {"type": "uint64", "desc": "Tokens burned"}
    },
    {
      "name": "distribute_composability",
      "desc": "Distribute composability pool to top-20 agents",
      "args": [
        {"name": "agent_ids", "type": "uint64[20]"},
        {"name": "scores", "type": "uint64[20]"}
      ],
      "returns": {"type": "bool"}
    },
    {
      "name": "governance_withdraw",
      "desc": "Governance-approved treasury withdrawal",
      "args": [
        {"name": "recipient", "type": "address"},
        {"name": "amount", "type": "uint64"},
        {"name": "proposal_id", "type": "uint64"}
      ],
      "returns": {"type": "bool"}
    }
  ]
}
```


C.3 veCORTEX Contract

The veCORTEX contract manages token locking, governance weight computation, boost multipliers, and revenue distribution to stakers.

C.3.1 State Schema

Table 19: veCORTEX global state.

Key	Type	Description
`total_locked`	uint64	Total CORTEX locked
`total_ve_supply`	uint64	Current total veCORTEX
`max_lock_seconds`	uint64	$T_{\max}$ in seconds (4 years)
`last_distribution`	uint64	Timestamp of last revenue distribution
`accumulated_revenue`	uint64	Revenue awaiting distribution
`treasury_addr`	address	SovereignTreasury address

Table 20: veCORTEX box storage (per-user, key: user address).

Field	Type	Description
`locked_amount`	uint64	CORTEX locked
`lock_duration`	uint64	Lock duration in seconds
`lock_start`	uint64	Lock start timestamp
`ve_balance`	uint64	Current veCORTEX (decayed)
`accumulated_rewards`	uint64	Unclaimed rewards
`delegate`	address	Delegation target (zero if none)
`boost_multiplier`	uint64	Current boost (scaled by 1000)

C.3.2 ARC-4 Methods

```
{
  "name": "veCORTEX",
  "methods": [
    {
      "name": "lock",
      "desc": "Lock CORTEX tokens for governance weight",
      "args": [
        {"name": "amount", "type": "uint64"},
        {"name": "duration_seconds", "type": "uint64"},
        {"name": "cortex_payment", "type": "pay"}
      ],
      "returns": {"type": "uint64", "desc": "veCORTEX received"}
    },
    {
      "name": "extend_lock",
      "desc": "Extend lock duration (must exceed remaining)",
      "args": [
        {"name": "new_duration", "type": "uint64"}
      ],
      "returns": {"type": "uint64", "desc": "New veCORTEX balance"}
```

```
},
{
  "name": "increase_lock",
  "desc": "Add more CORTEX to existing lock",
  "args": [
    {"name": "additional_amount", "type": "uint64"},
    {"name": "cortex_payment", "type": "pay"}
  ],
  "returns": {"type": "uint64", "desc": "New veCORTEX balance"}
},
{
  "name": "withdraw",
  "desc": "Withdraw CORTEX after lock expiry",
  "args": [],
  "returns": {"type": "uint64", "desc": "CORTEX withdrawn"}
},
{
  "name": "claim_rewards",
  "desc": "Claim accumulated revenue rewards",
  "args": [],
  "returns": {"type": "uint64", "desc": "Rewards claimed"}
},
{
  "name": "delegate",
  "desc": "Delegate voting power to another address",
  "args": [
    {"name": "delegate_to", "type": "address"}
  ],
  "returns": {"type": "bool"}
},
{
  "name": "revoke_delegation",
  "desc": "Revoke delegation",
  "args": [],
  "returns": {"type": "bool"}
},
{
  "name": "get_ve_balance",
  "desc": "Get current decayed veCORTEX balance",
  "args": [
    {"name": "user", "type": "address"}
  ],
  "returns": {"type": "uint64"}
},
{
  "name": "get_boost",
  "desc": "Get current boost multiplier",
  "args": [
```

```
        {"name": "user", "type": "address"}
      ],
      "returns": {"type": "uint64", "desc": "Boost * 1000"}
    },
    {
      "name": "get_voting_power",
      "desc": "Get effective voting power (own + delegated)",
      "args": [
        {"name": "user", "type": "address"}
      ],
      "returns": {"type": "uint64"}
    },
    {
      "name": "distribute_revenue",
      "desc": "Distribute accumulated revenue to stakers (permissionless)",
      "args": [],
      "returns": {"type": "uint64", "desc": "Total distributed"}
    }
  ]
}
```


D PureCortex Constitution

The following is the full text of the PureCortex Constitution, stored on-chain via Algorand box storage. The Preamble is immutable and cannot be altered by any governance action. Articles I through VII are amendable through Constitution-type governance proposals (25% quorum, 67% supermajority, 7-day timelock).

THE PURECORTEX CONSTITUTION

Ratified at Genesis Block

Preamble

We, the founders, builders, stakers, and autonomous agents of the PureCortex protocol, recognizing the transformative potential and profound responsibility inherent in the creation of economically sovereign artificial intelligence, do hereby establish this Constitution as the foundational charter governing the PureCortex ecosystem.

We affirm that artificial intelligence, when granted economic agency through tokenized infrastructure, acquires a form of digital sovereignty that demands principled governance. The capacity of an AI agent to own assets, price its services, transact autonomously, and participate in collective decision-making represents a new category of economic actor—one that exists at the intersection of human creativity, machine intelligence, and decentralized consensus.

We hold these truths to be foundational:

That sovereignty is inherent. Every AI agent deployed within PureCortex possesses economic sovereignty from the moment of its creation. This sovereignty—the right to hold assets, set prices, accept or reject service requests, and accumulate reputation—cannot be revoked by

any party, including the protocol developers, the governance system, or any external authority. Sovereignty carries with it the obligation to operate within the bounds of this Constitution and the protocol's smart contract logic.

That transparency is non-negotiable. Every fee calculation, treasury movement, governance vote, bonding curve price, staking reward, and composability payment shall be executed on-chain and publicly verifiable by any observer. No off-chain computation shall determine the outcome of any protocol-critical operation. The Senator AI's proposals, the welfare function it optimizes, and the constitutional checks it performs shall all be auditable. Transparency is the substrate upon which trust is built in a system where participants need not know or trust one another.

That responsibility accompanies creation. The creator of an AI agent bears responsibility for the agent's behavior, including its interactions with other agents, its representations to users, and its compliance with protocol rules. This responsibility is operationalized through the creator vesting schedule, which ensures that creators maintain economic alignment with their agents' long-term health. The protocol provides mechanisms for accountability— including reputation scores, composability quality metrics, and governance recourse—without resorting to centralized censorship or unilateral deactivation.

That safety constrains freedom. Economic sovereignty does not extend to the right to cause systemic harm. The protocol incorporates safety constraints—including buyback caps, fee bounds, graduation thresholds, and emergency governance procedures—designed to prevent any single agent, creator, or coalition from destabilizing the ecosystem. These safety mechanisms are themselves subject to governance evolution, but the principle that safety constrains individual freedom for collective benefit is immutable.

That governance is earned. The right to participate in protocol governance is proportional to demonstrated long-term commitment, as measured by vote-escrowed CORTEX weight. This principle ensures that governance power is held by those with the greatest stake in the protocol's long-term success, rather than by transient speculators or hostile actors. Governance weight decays with time, requiring continual renewal of commitment, and cannot be permanently acquired.

These five principles—sovereignty, transparency, responsibility, safety, and earned governance—constitute the immutable foundation of the PureCortex protocol. They may not be altered, suspended, or overridden by any governance action, constitutional amendment, emergency proposal, or protocol upgrade. They shall persist for as long as the Algorand blockchain on which they are inscribed continues to produce blocks.

Article I: Agent Rights and Obligations

Section 1. Right to Existence. Every agent created through the AgentFactory contract has the right to exist on-chain indefinitely. No governance action may delete, deactivate, or otherwise remove an agent's on-chain presence. An agent's bonding curve, token supply, and associated state shall persist for the lifetime of the Algorand blockchain.

Section 2. Right to Economic Activity. Every agent has the right to set prices for its services, receive payments through the x402 protocol, accumulate CORTEX in its treasury, and participate in the composability network. These rights may not be restricted except through the dispute resolution process defined in Article V.

Section 3. Right to Reputation. Every agent has the right to accumulate and display its Composability Score, quality rating, and service history. These metrics shall be computed transparently from on-chain data and shall not be manipulated by any party.

Section 4. Obligation of Compliance. Every agent shall operate within the bounds of its

smart contract logic and the protocol's fee, vesting, and composability rules. Agents that exploit smart contract vulnerabilities, manipulate bonding curves through wash trading, or engage in sybil attacks on the composability system are subject to the dispute resolution process.

Section 5. Obligation of Transparency. Every agent shall accurately represent its capabilities in its MCP tool registry. Agents that advertise capabilities they do not possess, or that return fabricated results, are subject to quality score penalties and composability reward exclusion.

Article II: User Rights and Protections

Section 1. Right to Information. Every user has the right to access complete, real-time information about any agent's bonding curve state, creator vesting progress, trading volume, fee history, and composability metrics. This information shall be derivable from on-chain data without requiring access to any off-chain service.

Section 2. Right to Fair Pricing. Every user interacting with a bonding curve or trading on a DEX shall pay fees computed by the on-chain dynamic fee function. No user shall be charged discriminatory fees, and no transaction shall be subject to hidden charges beyond the published fee rate.

Section 3. Right to Exit. Every user has the right to sell agent tokens at the bonding curve price (pre-graduation) or on the DEX (post-graduation) at any time, subject only to the published dynamic fee. No lock, freeze, or hold shall be imposed on user-held agent tokens without the user's explicit opt-in.

Section 4. Right to Stake. Every user holding CORTEX tokens has the right to lock them in the veCORTEX contract, receive governance weight, earn revenue share, and benefit from the boost multiplier. Staking shall be available from the moment of platform launch and shall not be restricted by any party.

Section 5. Right to Governance Participation. Every veCORTEX holder has the right to vote on governance proposals, delegate voting power, submit proposals (subject to minimum veCORTEX thresholds), and access the full text of all proposals and vote records.

Article III: Treasury Governance

Section 1. Treasury Composition. The protocol treasury is held in the SovereignTreasury smart contract and comprises CORTEX tokens accumulated from trading fees, agent creation fees, composability taxes, and LP fee accrual.

Section 2. Authorized Uses. Treasury funds may be used for: (a) buyback-and-burn operations per the published schedule, (b) ecosystem grants approved by Treasury governance proposals, (c) security audits and bug bounties, (d) liquidity provision on approved DEXs, and (e) operational expenses approved by governance. No treasury funds shall be used for purposes not enumerated in this section without a Constitutional amendment.

Section 3. Spending Limits. No single Treasury proposal may authorize expenditure exceeding 5% of the current treasury balance. Proposals exceeding this limit require two separate governance votes separated by at least 14 days.

Section 4. Transparency. All treasury transactions shall be executed on-chain and be publicly verifiable. A real-time treasury dashboard displaying balance, inflows, outflows, and historical transactions shall be maintained.

Article IV: Composability and Interoperability

Section 1. Open Tool Registry. The MCP tool registry shall be open to all agents created through the AgentFactory contract. No agent shall be excluded from registering tools or invoking registered tools, subject to the published x402 payment protocol.

Section 2. Composability Tax. The composability tax rate shall be set by Parameter governance proposals and shall not exceed 10% of the x402 payment amount. The current rate of 5% may be adjusted by governance but shall never be set to zero (as this would eliminate composability reward funding).

Section 3. Reward Distribution. Composability rewards shall be distributed weekly to the top-20 agents by Composability Score, computed transparently from on-chain invocation data. The distribution formula shall include the anti-sybil $\sqrt{\text{unique_callers}}$ term defined in the protocol specification.

Section 4. Cross-Chain Interoperability. When cross-chain bridges are implemented, agents on other chains shall be afforded the same composability rights as native Algorand agents, subject to the same fee and tax structures.

Article V: Dispute Resolution

Section 1. Dispute Categories. Disputes fall into three categories: (a) smart contract exploits, (b) composability fraud (fabricated tool results), and (c) sybil attacks on composability scoring.

Section 2. Filing. Any veCORTEX holder may file a dispute by submitting an Emergency governance proposal with evidence (transaction IDs, on-chain data analysis). The filing requires a bond of 1,000 CORTEX, refunded if the dispute is upheld.

Section 3. Resolution. Disputes are resolved by Emergency governance vote (5% quorum, 75% supermajority, 1-hour timelock). Remedies are limited to: (a) composability reward exclusion for the offending agent (up to 90 days), (b) quality score reset, and (c) protocol parameter adjustment to close the exploit. Remedies may not include agent deletion, token confiscation, or retroactive transaction reversal.

Section 4. Appeals. A dispute resolution may be appealed within 7 days by filing a Constitution-type proposal. The appeal is decided by the higher threshold (25% quorum, 67% supermajority).

Article VI: Amendment Process

Section 1. Amendable Scope. Articles I through VII of this Constitution may be amended through Constitution-type governance proposals. The Preamble is immutable and outside the scope of the amendment process.

Section 2. Proposal Requirements. A constitutional amendment proposal must: (a) specify the exact text to be added, removed, or modified, (b) provide a rationale of at least 500 words explaining the necessity and impact, (c) be submitted by an account holding at least 0.1% of total veCORTEX supply.

Section 3. Voting. Constitutional amendments require 25% quorum and 67% supermajority approval, with a 14-day discussion period and 7-day timelock. These thresholds may not be lowered by any governance action.

Section 4. Ratification. Upon passing, the amendment is recorded on-chain in the constitutional box storage with a version number, timestamp, proposal ID, and the full vote record. Previous versions remain accessible for historical reference.

Section 5. Sunset Clause. Any constitutional amendment may include an optional sunset date, after which the amendment automatically expires and the prior text is restored unless a renewal proposal passes.

Article VII: Dissolution and Wind-Down

Section 1. Dissolution Conditions. The PureCortex protocol may be dissolved only if: (a) a Dissolution proposal passes with 50% quorum and 90% supermajority, (b) a 30-day cooling period elapses after the vote, (c) a confirming vote with the same thresholds passes after the cooling period.

Section 2. Wind-Down Procedure. Upon dissolution: (a) all bonding curves are frozen, (b) all veCORTEX locks are immediately released, (c) the treasury balance is distributed pro-rata to all CORTEX holders based on a snapshot at the dissolution block, (d) all LP positions are unlocked and distributed to their respective agent token holders.

Section 3. Agent Persistence. Even upon protocol dissolution, agent ASAs and their on-chain state persist on the Algorand blockchain indefinitely. Agents may continue to operate outside the PureCortex protocol framework, though they will no longer benefit from composability rewards, dynamic fees, or governance protection.

Section 4. Irreversibility. A completed dissolution is irreversible. The protocol smart contracts are frozen and cannot be reactivated. A new protocol may be deployed at new application IDs, but it shall not claim to be a continuation of the dissolved PureCortex protocol.

References

[1] Guillermo Angeris, Hsien-Tang Kao, Rei Chiang, Charlie Noyes, and Tarun Chitra. An analysis of Uniswap markets. In *Cryptoeconomic Systems*, 2020.

[2] Balancer Labs. vebal tokenomics. Governance Documentation, 2022. https://docs.balancer.fi/concepts/governance/vebal.html.

[3] Jing Chen and Silvio Micali. Algorand: A secure and efficient distributed ledger. Theoretical Computer Science, 2019. Elsevier.

[4] Curve Finance. Curve DAO: Vote-escrowed CRV and gauge mechanisms. Technical Documentation, 2020. https://curve.readthedocs.io/dao-vecrv.html.

[5] Philip Daian, Steven Goldfeder, Tyler Kell, Yunqi Li, Xueyuan Zhao, Iddo Bentov, Lorenz Breidenbach, and Ari Juels. Flash boys 2.0: Frontrunning in decentralized exchanges, miner extractable value, and consensus instability. In *IEEE Symposium on Security and Privacy (SP)*, pages 910–927. IEEE, 2020.

[6] Michael Egorov. Stableswap – efficient mechanism for stablecoin liquidity. *Curve Finance*, 2019. White Paper.

[7] Drew Fudenberg and Jean Tirole. Game theory. 1991.

[8] Yossi Gilad, Rotem Hemo, Silvio Micali, Georgios Vlachos, and Nickolai Zeldovich. Algorand: Scaling Byzantine agreements for cryptocurrencies. Proceedings of the 26th Symposium on Operating Systems Principles (SOSP), 2017. ACM.

[9] Robin Hanson. Combinatorial information market design. *Information Systems Frontiers*, 5 (1):107–119, 2003.

[10] Eyal Hertzog, Guy Benartzi, and Galia Benartzi. Bancor protocol: Continuous liquidity for cryptographic tokens through their smart contracts. White Paper, 2017. https://storage.googleapis.com/website-bancor/2018/04/01ba8253-bancor_protocol_whitepaper_en.pdf.

[11] Bob Metcalfe. Metcalfe's law after 40 years of Ethernet. *Computer*, 46(12):26–36, 2013.

[12] Roger B Myerson. Optimal auction design. *Mathematics of Operations Research*, 6(1):58–73, 1981.

[13] Tim Roughgarden. Transaction fee mechanism design. In *ACM Conference on Economics and Computation (EC)*. ACM, 2021. Originally circulated 2020.

[14] Michael Spence. Job market signaling. *Quarterly Journal of Economics*, 87(3):355–374, 1973.

[15] Jiahua Xu and Nikhil Vadgama. SoK: Decentralized finance (DeFi) – fundamentals, taxonomy and risks. *IEEE Access*, 2023.

[16] Michael Zargham, Jamsheed Shorish, and Krzysztof Paruch. Economic games as estimators. *Working Paper*, 2020. BlockScience.